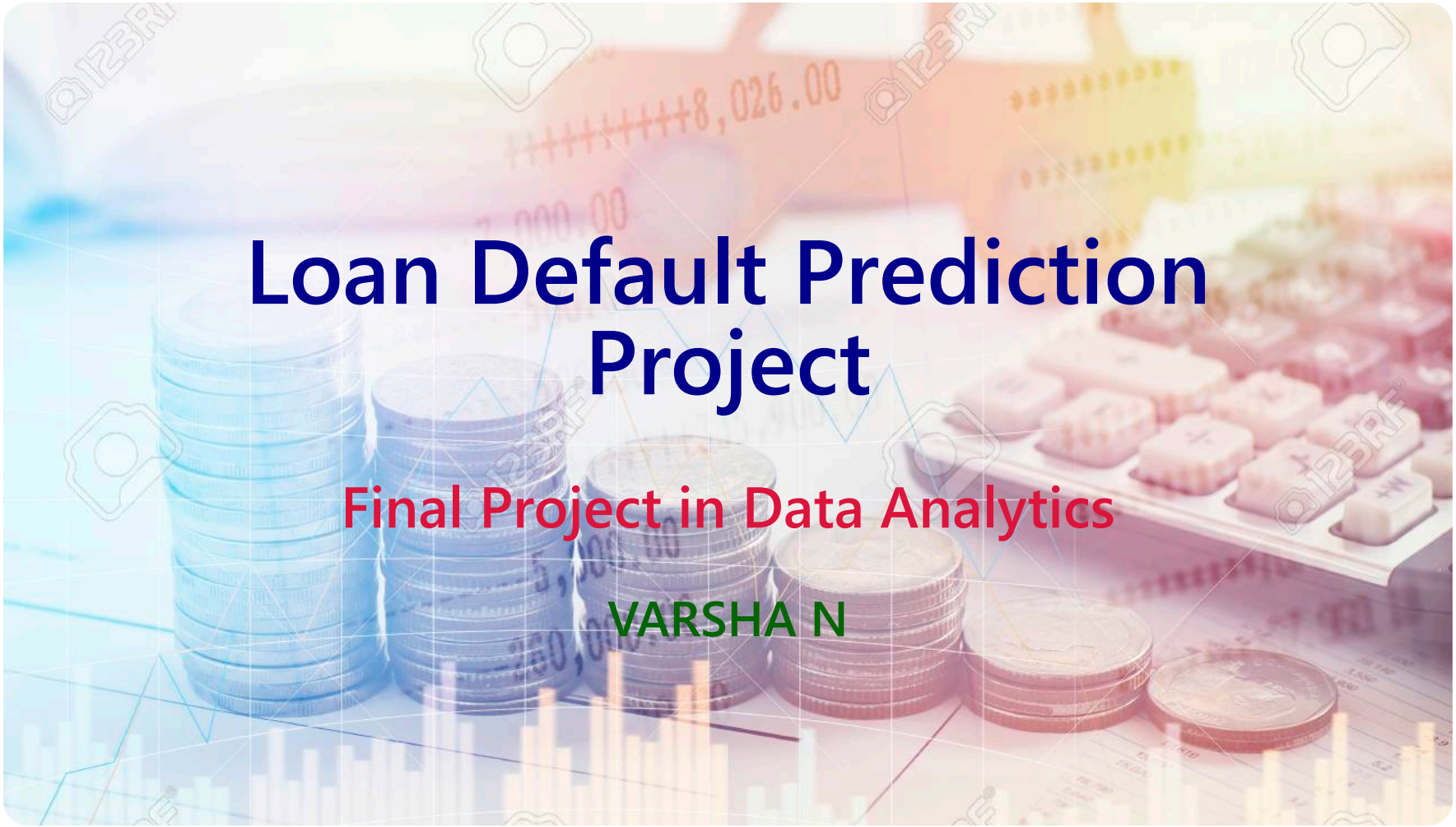


Loan Default Prediction Project

Final Project in Data Analytics

VARSHA N

Introduction

Loans are a major part of the banking system. However, many applicants default on their loans, causing financial losses for banks. Predicting loan default in advance can help financial institutions minimize risk and improve decision-making. With the help of data analytics, we can analyze borrower characteristics to predict the probability of loan default.

🔹 Aim

The aim of this project is to analyze loan applicant data and develop a predictive model that can determine the likelihood of a borrower defaulting on a loan.

- 📊 Understand borrower characteristics such as income, age, credit score, loan amount, and employment details.
- 🔍 Identify patterns and key factors that influence repayment or default.
- 🤖 Build a predictive model for loan default.
- 📈 Support banks in better lending decisions and risk reduction.

🔹 Dataset Description

- Source -> Kaagle
- Rows Count -> 322375
- Column Count -> 18

🔹 Column-wise Description

1. LoanID

- **Type:** Identifier(String)
- **Meaning:** A unique ID assigned to each loan record
- **Use in Analysis:**
  - Not used in prediction(it's just an identifier).

2. Age

- **Type:** Integer
- **Meaning:** Age of the Borrower.

- **Importance:**
  - Younger borrowers may have lower credit history and higher default risk.
  - Middle-aged borrowers may be more stable financially.
  - Very old borrowers may have limited income sources.

3. Income

- **Type:** Integer
- **Meaning:** Annual Income of the Borrower.
- **Importance:**
  - Higher Income usually means lower default risk(borrower can repay).
  - Very low income as a higher chance of dafault ,if loan amount is high.

4. LoanAmount

- **Type:** Integer
- **Meaning:** Amount of money borrowed.
- **Importance:**
  - High loan amount compared to income may increase risk.
  - Used along with income to calculate affordability ratios.

5. CreditSore

- **Type:** Integer
- **Meaning:**Credit score of the borrower(measure of credit worthiness).
- **Importance:**
  - Higher score,borrower is financially reliable.
  - Low score,higher chance of default.

6. MonthsEmployed

- **Type:** Integer
- **Meaning:** Number of months borrower has been employed.
- **Importance:**

- If the borrower has a long term employment then job is stable and has less risks.
- Unstable jobs or short term employment then default risk is higher.

7. NumCreditLines

- **Type:** Integer
- **Meaning:**Number of credit lines borrower has opened(credit cards,loans) .
- **Importance:**
  - If someone has too many credit lines,it means they are borrowing too much, higher chance of default.
  - If someone has very few or none,the bank cannot judge their repayment history well and it's harder to trust.

8. InterestRate

- **Type:** Float
- **Meaning:** Interest rate charged on loan.
- **Importance:**
  - Higher interest rates,high repayment burden.
  - Loans with high rates are riskier and often given to high-risk borrowers.

9. LoanTerm

- **Type:** Integer
- **Meaning:** Duration of the loan in months.
- **Importance:**
  - Short term loans,higher monthly payments but less overall risks.
  - Long term loans,lesser monthly payments but higher long term risks.

10. DTIRatio(Debt-to-Income Ratio)

- **Type:** Float
- **Meaning:** Ratio of borrower's debt to income.
- **Importance:**
  - Low DTI,Person has enough free income left and it's safer for banks.
  - High DTI,most income goes to paying existing debts and riskier to give loans.

- Moreover,DTI tells if the borrower can handle another loan or not.

11. Education

- **Type:** String(phD,Master's,Bachelor's,High School)
- **Meaning:** Highest level education of borrower.
- **Importance:**
  - Higher education often correlates with higher income potential.
  - Low education may limit future earning capacity.

12. EmploymentType

- **Type:** String ((Full-time,Part-time,Self-Employed,Unemployed)
- **Meaning:** Amount of money borrowed.
- **Importance:**
  - High loan amount compared to income may increase risk.
  - Used along with income to calculate affordability ratios.

13. MaritalStatus

- **Type:** String(Single,Married,Divorced)
- **Meaning:** Borrower's marital status.
- **Importance:**
  - Married borrowers may have shared finacial responsibilities(sometimes more stable).
  - Single borrowers may have fewer dependents.
  - Divorced borrowers may face financial stress.

14. HasMortage

- **Type:** String(Yes/No)
- **Meaning:** Whether borrower already has a mortgage.
- **Importance:**
  - If there is any existing mortgage, then it will be a larger debt and risk increases.
  - If nothing is there then lower debt burden.

15. HasDependents

- **Type:** String(Yes/No)
- **Meaning:** Whether borrower already has dependents(children/others to support).
- **Importance:**
  - Dependents increase expenses,repayment becomes harder.
  - If no dependents,borrower has more disposable income.

16. HasMortage

- **Type:** String(Yes/No)
- **Meaning:** Whether borrower already has a mortgage.
- **Importance:**
  - If there is any existing mortgage, then it will be a larger debt and risk increases.
  - If nothing is there then lower debt burden.

17. HasCoSigner

- **Type:** String(Yes/No)
- **Meaning:** Whether loan has a co-signer(another person guaranteeing repayment).
- **Importance:**
  - Co-signer reduces risk,it's their legal responsibility.
  - If no Co-signer,lender takes more risk.

18. Default

- **Type:** Integer(0/1)
- **Meaning:**
  - 0=Loan was repaid successfully(No Default)
  - 1= Borrower failed to repay(Default)
- **Importance:**
  - If there is any existing mortgage, then it will be a larger debt and risk increases.
  - If nothing is there then lower debt burden.

🔹 Project Scope

1. Data Loading and Intial Overview

➡️ Import the dataset using Pandas and provide an Overview

In [28]:
```
# Importing libraries
import pandas as pd
import numpy as np
df = pd.read_csv("C:\python1\Loan_default_uncleaned_2.csv")
```

-> Number of rows and columns

In [29]:
```
df.shape
```

Out[29]:
```
(322375, 18)
```

In [3]:
```
# To get only Rows
df.shape[0]
```

Out[3]:
```
322375
```

In [4]:
```
# To get only Columns
df.shape[1]
```

Out[4]:
```
18
```

In [5]:
```
# To get Column Names
df.columns
```

Out[5]:
```
Index(['LoanID', 'Age', 'Income', 'LoanAmount', 'CreditScore',
       'MonthsEmployed', 'NumCreditLines', 'InterestRate', 'LoanTerm',
       'DTIRatio', 'Education', 'EmploymentType', 'MaritalStatus',
       'HasMortgage', 'HasDependents', 'LoanPurpose', 'HasCoSigner',
       'Default'],
      dtype='object')
```

-> Data types of each column

In [6]:
```
df.dtypes
```

Out[6]:
```
LoanID             object
Age               float64
Income            float64
LoanAmount        float64
CreditScore       float64
MonthsEmployed      int64
NumCreditLines      int64
InterestRate      float64
LoanTerm            int64
DTIRatio          float64
Education          object
EmploymentType     object
MaritalStatus      object
HasMortgage        object
HasDependents      object
LoanPurpose        object
HasCoSigner        object
Default             int64
dtype: object
```

In [7]:
```
# To get datatype of single column
df["LoanAmount"].dtype
```

Out[7]:
```
dtype('float64')
```

-> Initial Observations

In [8]:
```
# HEAD()
```

```
df.head()   #Shows the first few rows of the dataset(default 5 rows)
```

Out[8]:

	LoanID	Age	Income	LoanAmount	CreditScore	MonthsEmployed	NumCreditLines	InterestRate	LoanTerm	DTIRatio	Educatio
0	I38PQUQS96	56.0	85994.0	50587.0	520.0	80	4	15.23	36	0.44	Bachelor'
1	HPSK72WA7R	69.0	50432.0	124440.0	458.0	15	1	4.81	60	0.68	Master'
2	C1OZ6DPJ8Y	46.0	84208.0	129188.0	451.0	26	3	21.17	24	0.31	Master'
3	V2KKSFM3UN	32.0	31713.0	44799.0	743.0	0	3	7.07	24	0.23	Hig Schoo
4	EY08JDHTZP	60.0	20437.0	9139.0	633.0	8	4	6.51	48	0.73	Bachelor'

In [9]:

```
df.head(10)   # Shows first 10 rows of the dataset
```

Out[9]:

	LoanID	Age	Income	LoanAmount	CreditScore	MonthsEmployed	NumCreditLines	InterestRate	LoanTerm	DTIRatio	Educati
0	I38PQUQS96	56.0	85994.0	50587.0	520.0	80	4	15.23	36	0.44	Bachelo
1	HPSK72WA7R	69.0	50432.0	124440.0	458.0	15	1	4.81	60	0.68	Maste
2	C1OZ6DPJ8Y	46.0	84208.0	129188.0	451.0	26	3	21.17	24	0.31	Maste
3	V2KKSFM3UN	32.0	31713.0	44799.0	743.0	0	3	7.07	24	0.23	Hi Scho
4	EY08JDHTZP	60.0	20437.0	9139.0	633.0	8	4	6.51	48	0.73	Bachelo
5	A9S62RQ7US	25.0	90298.0	90448.0	720.0	18	2	22.72	24	0.10	Hi Scho
6	H8GXPAOS71	38.0	111188.0	177025.0	429.0	80	1	19.11	12	0.16	Bachelo
7	0HGZQKJ36W	56.0	126802.0	155511.0	531.0	67	4	8.15	60	0.43	Ph
8	1R0N3LGNRJ	36.0	42053.0	92357.0	827.0	83	1	23.94	48	0.20	Bachelo
9	CM9L1GTT2P	40.0	132784.0	228510.0	480.0	114	4	9.09	48	0.33	Hi Scho

In [10]:

```
# TAIL()
df.tail()  #Shows last 5 default rows of the dataset
```

Out[10]:

	LoanID	Age	Income	LoanAmount	CreditScore	MonthsEmployed	NumCreditLines	InterestRate	LoanTerm	DTIRatio	Ec
322370	PYQE3BDKX3	NaN	NaN	NaN	NaN	94	3	2.88	24	0.82	M
322371	0MF69ZQSJB	NaN	NaN	NaN	NaN	4	2	19.60	12	0.63	
322372	7ISLSUD0G2	NaN	NaN	NaN	NaN	85	1	15.03	60	0.47	M
322373	FJ6QE5V305	NaN	NaN	NaN	NaN	43	4	17.56	12	0.54	
322374	NWANQE3AER	NaN	NaN	NaN	NaN	69	2	5.74	36	0.13	b

In [ ]:

```
df.tail(10)  #Shows last 10 rows of the dataset
```

In [11]:

```
# INDEX()
df.index  # Shows the row labels(index) of the dataframe
```

Out[11]:

```
RangeIndex(start=0, stop=322375, step=1)
```

In [12]:

```
# INFO()
# Number of rows
# Column names
# Data Types
# Non-null counts
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 322375 entries, 0 to 322374
Data columns (total 18 columns):
 #   Column          Non-Null Count   Dtype
---  ------          --------------   -----
 0   LoanID          322375 non-null  object
 1   Age             312799 non-null  float64
 2   Income          312799 non-null  float64
 3   LoanAmount      312799 non-null  float64
 4   CreditScore     312799 non-null  float64
 5   MonthsEmployed  322375 non-null  int64
 6   NumCreditLines  322375 non-null  int64
 7   InterestRate    322375 non-null  float64
 8   LoanTerm        322375 non-null  int64
 9   DTIRatio        322375 non-null  float64
 10  Education       322375 non-null  object
 11  EmploymentType  322375 non-null  object
 12  MaritalStatus   322375 non-null  object
 13  HasMortgage     322375 non-null  object
 14  HasDependents   322375 non-null  object
 15  LoanPurpose     322375 non-null  object
 16  HasCoSigner     322375 non-null  object
 17  Default         322375 non-null  int64
dtypes: float64(6), int64(4), object(8)
memory usage: 44.3+ MB
```

In [13]:

```python
# DESCRIBE()
# Provides statistical summary for numeric columns
# Count,Mean,Std,Min,25%,50%,75%,Max
df.describe()   # Numerical Columns
```

Out[13]:

	Age	Income	LoanAmount	CreditScore	MonthsEmployed	NumCreditLines	InterestRate	LoanTerm	
count	312799.000000	312799.000000	312799.000000	312799.000000	322375.000000	322375.000000	322375.000000	322375.000000	322
mean	43.500011	82492.467060	127598.987104	574.378064	59.567432	2.500278	13.497270	36.036144	
std	14.994360	38961.793881	70837.575367	158.864188	34.640502	1.116941	6.635982	16.987138	
min	18.000000	15000.000000	5000.000000	300.000000	0.000000	1.000000	2.000000	12.000000	
25%	31.000000	48811.000000	66190.500000	437.000000	30.000000	2.000000	7.780000	24.000000	
50%	43.000000	82484.000000	127506.000000	574.000000	60.000000	2.000000	13.460000	36.000000	
75%	56.000000	116191.500000	189062.500000	712.000000	90.000000	3.000000	19.260000	48.000000	
max	69.000000	149999.000000	249999.000000	849.000000	119.000000	4.000000	25.000000	60.000000	

In [14]:

```
df.describe(include='all')    # All Columns
```

Out[14]:

	LoanID	Age	Income	LoanAmount	CreditScore	MonthsEmployed	NumCreditLines	InterestRate	
count	322375	312799.000000	312799.000000	312799.000000	312799.000000	322375.000000	322375.000000	322375.000000	322
unique	255347	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
top	26XYW14YHT	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
freq	3	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
mean	NaN	43.500011	82492.467060	127598.987104	574.378064	59.567432	2.500278	13.497270	
std	NaN	14.994360	38961.793881	70837.575367	158.864188	34.640502	1.116941	6.635982	
min	NaN	18.000000	15000.000000	5000.000000	300.000000	0.000000	1.000000	2.000000	
25%	NaN	31.000000	48811.000000	66190.500000	437.000000	30.000000	2.000000	7.780000	
50%	NaN	43.000000	82484.000000	127506.000000	574.000000	60.000000	2.000000	13.460000	
75%	NaN	56.000000	116191.500000	189062.500000	712.000000	90.000000	3.000000	19.260000	
max	NaN	69.000000	149999.000000	249999.000000	849.000000	119.000000	4.000000	25.000000	

2. Data Pre-Processing

➡️ Perform all neccessary cleaning steps such as :

➡️ Handling Missing Values

In [15]:
```
#Count missing values of each columns
df.isnull().sum()
```

Out[15]:
```
LoanID               0
Age               9576
Income            9576
LoanAmount        9576
CreditScore       9576
MonthsEmployed       0
NumCreditLines       0
InterestRate         0
LoanTerm             0
DTIRatio             0
Education            0
EmploymentType       0
MaritalStatus        0
HasMortgage          0
HasDependents        0
LoanPurpose          0
HasCoSigner          0
Default              0
dtype: int64
```

In [17]:
```
#Filling missing values
# -> Missing values in LoanAmount
#Median is the best choice as loan amounts can vary a lot.Median gives a fair middle value.
# ->Missing values in CreditScore
# people's scores are not spread evenly(most people are average,only a few have very low or very high).So we use median.
# ->Missing values in Income
# Incomes are usually uneven, only a few people earn a lot more than others.Using median represents a typical income.
# ->Missing values in Age
# Most people are in a normal age(20-60),but if there are a few children or very old .Using median keeps the age correctly.
import warnings
# Ignore all warnings
warnings.filterwarnings("ignore")

df["Age"].fillna(df["Age"].median(), inplace=True)
df["Income"].fillna(df["Income"].median(), inplace=True)
df["LoanAmount"].fillna(df["LoanAmount"].median(), inplace=True)
df["CreditScore"].fillna(df["CreditScore"].median(), inplace=True)
df.tail(55)
```

Out[17]:

	LoanID	Age	Income	LoanAmount	CreditScore	MonthsEmployed	NumCreditLines	InterestRate	LoanTerm	DTIRatio	E
322320	QQVV5KC2G5	43.0	82484.0	127506.0	574.0	65	3	4.44	12	0.73	b
322321	A59MTVVARI	43.0	82484.0	127506.0	574.0	52	2	21.67	24	0.72	b
322322	BB67I4RKLN	43.0	82484.0	127506.0	574.0	2	1	8.29	60	0.30	N
322323	E9O7GNC8TE	43.0	82484.0	127506.0	574.0	112	3	6.90	12	0.46	N
322324	BW8206KOSJ	43.0	82484.0	127506.0	574.0	76	4	3.52	24	0.29	b
322325	ZE2YUDK21G	43.0	82484.0	127506.0	574.0	8	3	24.50	12	0.62	b
322326	OHZAKSMZBW	43.0	82484.0	127506.0	574.0	59	3	12.13	12	0.31	
322327	LDDMTHQW7M	43.0	82484.0	127506.0	574.0	107	1	24.85	60	0.15	
322328	NUG31RTOU0	43.0	82484.0	127506.0	574.0	74	2	24.58	12	0.74	N
322329	IMQO8C8K57	43.0	82484.0	127506.0	574.0	1	4	11.36	24	0.69	N
322330	PVJEONSUIH	43.0	82484.0	127506.0	574.0	106	2	11.65	36	0.67	
322331	903OT2UQPI	43.0	82484.0	127506.0	574.0	117	4	12.92	48	0.49	b
322332	8926FZRIQ8	43.0	82484.0	127506.0	574.0	101	3	19.67	60	0.38	
322333	WLOMJZMA5P	43.0	82484.0	127506.0	574.0	13	3	20.83	24	0.83	N
322334	7MLU510SSQ	43.0	82484.0	127506.0	574.0	106	3	2.51	48	0.26	N
322335	QLA01DFD76	43.0	82484.0	127506.0	574.0	96	4	12.83	36	0.53	
322336	UL8Z7TQOJV	43.0	82484.0	127506.0	574.0	100	1	3.06	24	0.22	
322337	843Z1BY8NI	43.0	82484.0	127506.0	574.0	97	2	10.53	48	0.38	
322338	XYMXOWZBNM	43.0	82484.0	127506.0	574.0	75	1	6.31	36	0.28	

	LoanID	Age	Income	LoanAmount	CreditScore	MonthsEmployed	NumCreditLines	InterestRate	LoanTerm	DTIRatio	E
322339	W1BPGEIRAZ	43.0	82484.0	127506.0	574.0	118	2	4.66	24	0.70	N
322340	L1LHJ4XIGW	43.0	82484.0	127506.0	574.0	32	3	16.87	12	0.26	
322341	SJZ1CQORGD	43.0	82484.0	127506.0	574.0	90	3	13.43	24	0.23	
322342	P6GGW4XVDU	43.0	82484.0	127506.0	574.0	33	3	19.98	48	0.29	b
322343	XL2GH7B5T1	43.0	82484.0	127506.0	574.0	28	3	14.57	24	0.33	N
322344	8MMH2KDN6J	43.0	82484.0	127506.0	574.0	65	3	15.43	24	0.17	b
322345	GFS3R8GENH	43.0	82484.0	127506.0	574.0	111	3	18.88	24	0.15	b
322346	II35MQ6ZYA	43.0	82484.0	127506.0	574.0	75	1	7.34	36	0.62	
322347	11IBUB4UYI	43.0	82484.0	127506.0	574.0	6	3	2.94	60	0.26	
322348	9FWEI1DYWZ	43.0	82484.0	127506.0	574.0	77	3	8.79	24	0.12	N
322349	GQ5UJRZZ9H	43.0	82484.0	127506.0	574.0	60	2	10.48	24	0.33	N
322350	IY9MMAP1E4	43.0	82484.0	127506.0	574.0	42	3	24.99	60	0.72	N
322351	3GATXJRKU9	43.0	82484.0	127506.0	574.0	113	4	10.44	24	0.62	b
322352	5ORYQ2EG5Y	43.0	82484.0	127506.0	574.0	106	4	5.05	36	0.81	
322353	USD54KQRQ7	43.0	82484.0	127506.0	574.0	63	1	14.78	48	0.67	
322354	JV3M5ES6WY	43.0	82484.0	127506.0	574.0	70	1	4.72	12	0.42	
322355	8YNLV79BC4	43.0	82484.0	127506.0	574.0	14	4	8.53	60	0.29	
322356	3CWMCCZZIK	43.0	82484.0	127506.0	574.0	84	1	10.37	48	0.48	b
322357	7PI6DB1N4M	43.0	82484.0	127506.0	574.0	119	4	9.20	36	0.55	

	LoanID	Age	Income	LoanAmount	CreditScore	MonthsEmployed	NumCreditLines	InterestRate	LoanTerm	DTIRatio	E
322358	T21EXWIKTY	43.0	82484.0	127506.0	574.0	115	2	22.11	24	0.82	
322359	EAFUROBVPQ	43.0	82484.0	127506.0	574.0	38	3	3.24	60	0.22	N
322360	X4TUGNIET0	43.0	82484.0	127506.0	574.0	39	2	13.30	36	0.17	
322361	94UEE939YF	43.0	82484.0	127506.0	574.0	76	4	10.97	12	0.65	N
322362	W61H219UDU	43.0	82484.0	127506.0	574.0	8	2	2.07	60	0.19	b
322363	J7CBWFLXFN	43.0	82484.0	127506.0	574.0	73	2	21.08	24	0.37	
322364	1IVWOLUAU4	43.0	82484.0	127506.0	574.0	24	1	12.38	36	0.15	
322365	YDZLFA17NO	43.0	82484.0	127506.0	574.0	108	3	5.74	60	0.19	N
322366	TQY19MUA96	43.0	82484.0	127506.0	574.0	82	4	6.99	12	0.73	
322367	CQ35I264AJ	43.0	82484.0	127506.0	574.0	66	1	16.19	24	0.43	b
322368	WNB79DK668	43.0	82484.0	127506.0	574.0	57	2	23.01	60	0.10	N
322369	CFJ1ZTVHVV	43.0	82484.0	127506.0	574.0	7	3	20.63	12	0.69	b
322370	PYQE3BDKX3	43.0	82484.0	127506.0	574.0	94	3	2.88	24	0.82	N
322371	0MF69ZQSJB	43.0	82484.0	127506.0	574.0	4	2	19.60	12	0.63	
322372	7ISLSUD0G2	43.0	82484.0	127506.0	574.0	85	1	15.03	60	0.47	N
322373	FJ6QE5V305	43.0	82484.0	127506.0	574.0	43	4	17.56	12	0.54	
322374	NWANQE3AER	43.0	82484.0	127506.0	574.0	69	2	5.74	36	0.13	b

In [18]:
```
# Verifying if missing values are filled
print(df.isnull().sum())
```

```
LoanID            0
Age               0
Income            0
LoanAmount        0
CreditScore       0
MonthsEmployed    0
NumCreditLines    0
InterestRate      0
LoanTerm          0
DTIRatio          0
Education         0
EmploymentType    0
MaritalStatus     0
HasMortgage       0
HasDependents     0
LoanPurpose       0
HasCoSigner       0
Default           0
dtype: int64
```

In [19]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 322375 entries, 0 to 322374
Data columns (total 18 columns):
 #   Column          Non-Null Count   Dtype
---  ------          --------------   -----
 0   LoanID          322375 non-null  object
 1   Age             322375 non-null  float64
 2   Income          322375 non-null  float64
 3   LoanAmount      322375 non-null  float64
 4   CreditScore     322375 non-null  float64
 5   MonthsEmployed  322375 non-null  int64
 6   NumCreditLines  322375 non-null  int64
 7   InterestRate    322375 non-null  float64
 8   LoanTerm        322375 non-null  int64
 9   DTIRatio        322375 non-null  float64
 10  Education       322375 non-null  object
 11  EmploymentType  322375 non-null  object
 12  MaritalStatus   322375 non-null  object
 13  HasMortgage     322375 non-null  object
 14  HasDependents   322375 non-null  object
 15  LoanPurpose     322375 non-null  object
 16  HasCoSigner     322375 non-null  object
 17  Default         322375 non-null  int64
dtypes: float64(6), int64(4), object(8)
memory usage: 44.3+ MB
```

✅ Removing Multiple Spaces, Leading /Trailing Spaces(Stripping),Changing to Proper Cases

In [7]:

```
#Convert to proper case and fix apostrophes
#categorical_cols = ["Education", "EmploymentType", "MaritalStatus", "LoanPurpose"],We create a list of column names that are
#.apply(lambda col: ...).Applies a function to each column in that selection.
# inside the lambda function (applied step by step):
#.astype(str)-Converts all values to string ( sometimes it might contain NaN, numbers, or mixed types).
#str.strip()-Removes leading and trailing spaces.
#.str.replace(r"[^a-zA-Z\s']", "", regex=True)-Removes all characters that are not letters, spaces, or apostrophes.
#.str.replace(r"\s+", " ", regex=True)-Replaces multiple spaces with a single space.
#.str.title()-Converts the string to Title Case (first letter capitalized, rest lowercase).
categorical_cols = ["Education", "EmploymentType", "MaritalStatus", "LoanPurpose"]
df[categorical_cols] = df[categorical_cols].apply(
    lambda col: col.astype(str)
```

```
                    .str.strip()                              # remove leading/trailing spaces
                    .str.replace(r"[^a-zA-Z\s']", "", regex=True) # keep only alphabets & '
                    .str.replace(r"\s+", " ", regex=True)     # collapse multiple spaces
                    .str.title()                              # proper case
)
print(df[categorical_cols].tail(15))
```

```
         Education EmploymentType MaritalStatus LoanPurpose
322360         Phd       Parttime       Married        Auto
322361    Master'S       Parttime      Divorced        Home
322362  Bachelor'S     Unemployed        Single        Home
322363 High School   Selfemployed       Married        Home
322364 High School   Selfemployed      Divorced       Other
322365    Master'S     Unemployed       Married   Education
322366         Phd   Selfemployed        Single        Home
322367  Bachelor'S       Parttime      Divorced       Other
322368    Master'S       Fulltime       Married       Other
322369  Bachelor'S       Parttime       Married        Home
322370    Master'S       Fulltime       Married       Other
322371 High School       Fulltime      Divorced   Education
322372    Master'S     Unemployed      Divorced        Auto
322373         Phd       Parttime        Single   Education
322374  Bachelor'S   Selfemployed       Married        Home
```

In [4]:
```
# Remove hash symbols from LoanPurpose column
df['LoanPurpose'] = df['LoanPurpose'].astype(str).str.replace('#', '', regex=False)

# Check again if hash symbol exists in LoanPurpose column
hash_exists = df['LoanPurpose'].str.contains('#').any()

print("Does '#' still exist in LoanPurpose?:", hash_exists)
```

```
Does '#' still exist in LoanPurpose?: False
```

In [5]:
```
# Clean 'LoanPurpose' column
df['LoanPurpose'] = df['LoanPurpose'].str.strip()       # remove spaces
df['LoanPurpose'] = df['LoanPurpose'].str.lower()       # make lowercase
```

In [6]:
```
df['LoanPurpose'].unique()
```

Out[6]: array(['other', 'auto', 'business', 'home', 'education'], dtype=object)

In [8]:
```
# Convert all values in Education column to proper case (first letter capitalized)
df["Education"] = df["Education"].str.strip().str.title()
```

In [9]:
```
print(df["Education"].unique())
```

```
["Bachelor'S" "Master'S" 'High School' 'Phd']
```

In [26]:
```
# Convert all values in EmploymentType column to proper case (first letter capitalized)
df["EmploymentType"] = df["EmploymentType"].str.strip().str.title()
```

In [27]:
```
print(df["EmploymentType"].unique())
```

```
['Fulltime' 'Unemployed' 'Selfemployed' 'Parttime']
```

In [28]:
```
# Convert all values in MaritalStatus column to proper case (first letter capitalized)
df["MaritalStatus"] = df["MaritalStatus"].str.strip().str.title()
```

In [29]:
```
print(df["MaritalStatus"].unique())
```

```
['Divorced' 'Married' 'Single']
```

In [30]:
```
#Checking if any row as null values
print(df[df.isnull().any(axis=1)])
```

```
Empty DataFrame
Columns: [LoanID, Age, Income, LoanAmount, CreditScore, MonthsEmployed, NumCreditLines, InterestRate, LoanTerm, DTIRatio, Educa
tion, EmploymentType, MaritalStatus, HasMortgage, HasDependents, LoanPurpose, HasCoSigner, Default]
Index: []
```

In [31]:
```
#Checking if all values are null or  not
print(df.isnull().sum())   # should show 0 for all columns(it shows null values are not there)
print(df.isnull().any().any())  # should return False( returns true if null values exists or false if no null values)
```

```
LoanID            0
Age               0
Income            0
LoanAmount        0
CreditScore       0
MonthsEmployed    0
NumCreditLines    0
InterestRate      0
LoanTerm          0
DTIRatio          0
Education         0
EmploymentType    0
MaritalStatus     0
HasMortgage       0
HasDependents     0
LoanPurpose       0
HasCoSigner       0
Default           0
dtype: int64
False
```

➡️ Removing Duplicates

In [32]:
```
# Count the number of duplicated rows in dataset
df.duplicated().sum()
```

Out[32]:
```
np.int64(60644)
```

In [34]:
```
#Removing duplicates across the dataset
df = df.drop_duplicates()
```

In [35]:
```
#After removing duplicates the total number of rows.
print(df.shape)
```

```
(261731, 18)
```

➡️ Correcting Data Types

In [36]:
```
#Correcting datatypes makes the dataset clean, efficient, and ready for analysis.
# Use of astype()- function is used to change the data type of a column (or entire DataFrame).

# Making a copy of the original  dataframe df.
df_cleaned = df.copy()

# 1. Converting Yes/No columns to boolean of HasDependents,HasCosigner
bool_columns = ["HasDependents", "HasCoSigner"]
for col in bool_columns:
    df_cleaned[col] = df_cleaned[col].map({"Yes": True, "No": False})

# 2. Converting Yes/No columns to boolean of HasMortgage
    df_cleaned["HasMortgage"] = df_cleaned["HasMortgage"].astype(bool)


# 3. Converting categorical text columns to category dtype
categorical_columns = ["Education", "EmploymentType", "MaritalStatus", "LoanPurpose"]
for col in categorical_columns:
    df_cleaned[col] = df_cleaned[col].astype("category")

# 4. Converting Default column to boolean (if needed otherwise keep it as integer)
df_cleaned["Default"] = df_cleaned["Default"].astype(bool)

# 5. Converting Age to nullable integer (inplace of float)
df["Age"] = df["Age"].astype("Int64")


#  Now checking the updated datatypes
print(df_cleaned.dtypes)
```

```
LoanID              object
Age                float64
Income             float64
LoanAmount         float64
CreditScore        float64
MonthsEmployed       int64
NumCreditLines       int64
InterestRate       float64
LoanTerm             int64
DTIRatio           float64
Education         category
EmploymentType    category
MaritalStatus     category
HasMortgage           bool
HasDependents         bool
LoanPurpose       category
HasCoSigner           bool
Default               bool
dtype: object
```

➡️ Creating derived columns

In [ ]:
```
# Derived columns  created from existing columns in your dataset.
# Their main purpose is to extract more meaningful information or make patterns easier to detect for analysis.
```

Loan_to_Income

In [37]:
```
# how much of loan amount compared to annual income
#if someone earns ₹10,00,000 annually and takes a loan of ₹5,00,000 → Loan_to_Income = 0.5 (50%).
#  Higher ratio means loan is big compared to income, which is riskier.
df['Loan_to_Income'] = df['LoanAmount'] / df['Income']
```

Credit_Utilization_Score

In [38]:
```
#loan utilization based on credit lines.
#Measures how much loan per credit line the borrower is using.
#+1 is added to avoid dividing by zero when someone has no credit lines.
#If a person has only 1-2 credit lines and high loan amount → their utilization is high, which increases risk.
df['Credit_Utilization'] = df['LoanAmount']/(df['NumCreditLines']+1)
```

Years_Employed

In [39]:
```
#Converted months of employment into years.
#Longer employment history usually indicates stability and lower risk.
df['Years_Employed'] = df['MonthsEmployed']/12
```

Age_Group

In [40]:
```
#Categorized customers(ypung,mid-age,Mature,Senior)
#It creates a new column called AgeGroup in your dataframe by grouping ages into categories (bins).
#pd.cut() → A pandas function used to segment continuous values into ranges (bins).
#bins=[18, 30, 45, 60, 100] → Defines the cut points (intervals) for the bins. ie,18 to 30,30 to 45,45 to 60,60 to 100
#labels=['Young', 'Mid-age', 'Senior', 'Elder'] → Names for these bins.
df['Age_Group'] = pd.cut(df['Age'], bins=[18, 30, 45, 60, 100],
                         labels=['Young', 'Mid-age', 'Mature', 'Senior'])
```

Risk_Score

In [41]:
```
#df['CreditScore']/850,Credit scores are usually between 300–850,Dividing by 850 normalizes the score to a range between 0 and
#- df['DTIRatio'],Debt-to-Income ratio = debt obligations / income.Higher DTI = more risky, so subtracting this reduces the sc
#- df['Loan_to_Income'],Loan amount relative to income,Larger loan burden compared to income increases risk, so it is subtract
#Custom score: higher= safer,lower=riskier(based on CreditScore,DTI,Loan_to_Income.)
df['Risk_Score'] = (df['Risk_Score']) = (df['CreditScore']/850) - df['DTIRatio'] - df['Loan_to_Income']
```

Has_Family_Dependents

In [42]:
```
#1 if borrower has dependents & not single,else 0
#derived from MaritalStatus and Dependents
#df['MaritalStatus'] != 'Single'.Checks if the borrower’s marital status is not single.Returns True if married/divorced/widowe
#df['HasDependents'] == 'Yes',Checks if the borrower has dependents.Returns True if dependents exist, otherwise False.
#& (logical AND)->Combines both conditions:Borrower is not single,Borrower has dependents.Both must be true for the final resu
#.astype(int)->Converts True/False into 1/0.
#Final result stored in new column->A new column HasFamilyDependents is created with:1 if borrower has dependents and is not s
df['HasFamilyDependents'] = ((df['MaritalStatus'] != 'Single') & (df['HasDependents'] == 'Yes')).astype(int)
```

In [43]:
```
# To see the whole (maximum number of columns) for displaying the output
pd.set_option('display.max_columns', None)
df.head()   # now all columns will be displayed
```

Out[43]:

	LoanID	Age	Income	LoanAmount	CreditScore	MonthsEmployed	NumCreditLines	InterestRate	LoanTerm	DTIRatio	Educatio
0	I38PQUQS96	56	85994.0	50587.0	520.0	80	4	15.23	36	0.44	Bachelor'
1	HPSK72WA7R	69	50432.0	124440.0	458.0	15	1	4.81	60	0.68	Master'
2	C1OZ6DPJ8Y	46	84208.0	129188.0	451.0	26	3	21.17	24	0.31	Master'
3	V2KKSFM3UN	32	31713.0	44799.0	743.0	0	3	7.07	24	0.23	Hig Schoo
4	EY08JDHTZP	60	20437.0	9139.0	633.0	8	4	6.51	48	0.73	Bachelor'

➡️ Filtering Data

In [ ]:
```
#Filtering means selecting only the rows that meet certain conditions from your dataset
```

High Income Borrowers

In [44]:
```
# Borrowers with Income > 50000
high_income = df[df['Income'] > 50000]
#print(high_income)          # shows all rows
print(high_income.head(10))
```

```
        LoanID  Age    Income  LoanAmount  CreditScore  MonthsEmployed  \
0   I38PQUQS96   56   85994.0     50587.0        520.0              80
1   HPSK72WA7R   69   50432.0    124440.0        458.0              15
2   C1OZ6DPJ8Y   46   84208.0    129188.0        451.0              26
5   A9S62RQ7US   25   90298.0     90448.0        720.0              18
6   H8GXPAOS71   38  111188.0    177025.0        429.0              80
7   0HGZQKJ36W   56  126802.0    155511.0        531.0              67
9   CM9L1GTT2P   40  132784.0    228510.0        480.0             114
10  IA35XVH6ZO   28  140466.0    163781.0        652.0              94
11  Y8UETC3LSG   28  149227.0    139759.0        375.0              56
13  GX5YQOGROM   53  117550.0     95744.0        395.0             112

    NumCreditLines  InterestRate  LoanTerm  DTIRatio    Education  \
0                4         15.23        36      0.44   Bachelor'S
1                1          4.81        60      0.68     Master'S
2                3         21.17        24      0.31     Master'S
5                2         22.72        24      0.10  High School
6                1         19.11        12      0.16   Bachelor'S
7                4          8.15        60      0.43          Phd
9                4          9.09        48      0.33  High School
10               2          9.08        48      0.23  High School
11               3          5.84        36      0.80          Phd
13               4          3.58        24      0.73  High School

   EmploymentType MaritalStatus HasMortgage HasDependents LoanPurpose  \
0        Fulltime      Divorced         Yes           Yes       other
1        Fulltime       Married          No            No       other
2      Unemployed      Divorced         Yes           Yes        auto
5      Unemployed        Single         Yes            No    business
6      Unemployed        Single         Yes            No        home
7        Fulltime       Married          No            No        home
9    Selfemployed       Married         Yes            No       other
10     Unemployed       Married          No            No   education
11       Fulltime      Divorced          No            No   education
13     Unemployed        Single          No            No        auto

   HasCoSigner  Default  Loan_to_Income  Credit_Utilization  Years_Employed  \
0          Yes        0        0.588262        10117.400000        6.666667
1          Yes        0        2.467481        62220.000000        1.250000
2           No        1        1.534154        32297.000000        2.166667
5          Yes        1        1.001661        30149.333333        1.500000
```

```
6          Yes        0        1.592123        88512.500000        6.666667
7          Yes        0        1.226408        31102.200000        5.583333
9          Yes        0        1.720915        45702.000000        9.500000
10          No        0        1.165983        54593.666667        7.833333
11         Yes        1        0.936553        34939.750000        4.666667
13         Yes        0        0.814496        19148.800000        9.333333

   Age_Group  Risk_Score  HasFamilyDependents
0     Mature   -0.416497                    1
1     Senior   -2.608657                    0
2     Mature   -1.313565                    1
5      Young   -0.254602                    0
6    Mid-age   -1.247417                    0
7     Mature   -1.031702                    0
9    Mid-age   -1.486209                    0
10     Young   -0.628924                    0
11     Young   -1.295377                    0
13    Mature   -1.079790                    0
```

Risky Customers

In [45]:
```
risky_customers = df[df['CreditScore'] < 600]
print(risky_customers.head(15))
```

```
        LoanID  Age    Income  LoanAmount  CreditScore  MonthsEmployed  \
0   I38PQUQS96   56   85994.0     50587.0        520.0              80
1   HPSK72WA7R   69   50432.0    124440.0        458.0              15
2   C1OZ6DPJ8Y   46   84208.0    129188.0        451.0              26
6   H8GXPAOS71   38  111188.0    177025.0        429.0              80
7   0HGZQKJ36W   56  126802.0    155511.0        531.0              67
9   CM9L1GTT2P   40  132784.0    228510.0        480.0             114
11  Y8UETC3LSG   28  149227.0    139759.0        375.0              56
13  GX5YQOGROM   53  117550.0     95744.0        395.0             112
15  O5DM5MPPNA   41   74064.0    230883.0        432.0              31
16  ZDDRGVTEXS   20  119704.0     25697.0        313.0              49
18  O1IKKLC69B   19   40718.0     78515.0        319.0             119
19  F7487UU2BF   41  123419.0    161146.0        376.0              65
20  7ASF0IHRIT   61   30142.0    133714.0        429.0              96
21  A22KI1B6SE   47  146113.0    100621.0        419.0              55
22  1MUSHWD9TW   55  132058.0    130912.0        583.0              48

    NumCreditLines  InterestRate  LoanTerm  DTIRatio    Education  \
0                4         15.23        36      0.44   Bachelor'S
1                1          4.81        60      0.68     Master'S
2                3         21.17        24      0.31     Master'S
6                1         19.11        12      0.16   Bachelor'S
7                4          8.15        60      0.43          Phd
9                4          9.09        48      0.33  High School
11               3          5.84        36      0.80          Phd
13               4          3.58        24      0.73  High School
15               2          5.00        60      0.89     Master'S
16               1          9.63        24      0.28          Phd
18               2         14.00        24      0.17   Bachelor'S
19               4         16.96        60      0.39  High School
20               1         15.58        12      0.65          Phd
21               1          9.32        12      0.38   Bachelor'S
22               4          5.82        60      0.47  High School

   EmploymentType MaritalStatus HasMortgage HasDependents LoanPurpose  \
0        Fulltime      Divorced         Yes           Yes       other
1        Fulltime       Married          No            No       other
2      Unemployed      Divorced         Yes           Yes        auto
6      Unemployed        Single         Yes            No        home
7        Fulltime       Married          No            No        home
9    Selfemployed       Married         Yes            No       other
```

11	Fulltime	Divorced	No	No	education
13	Unemployed	Single	No	No	auto
15	Unemployed	Married	Yes	No	auto
16	Unemployed	Single	Yes	No	home
18	Selfemployed	Divorced	Yes	No	education
19	Selfemployed	Single	Yes	No	other
20	Parttime	Divorced	No	Yes	business
21	Unemployed	Married	Yes	Yes	business
22	Unemployed	Married	No	Yes	business

	HasCoSigner	Default	Loan_to_Income	Credit_Utilization	Years_Employed	\
0	Yes	0	0.588262	10117.400000	6.666667	
1	Yes	0	2.467481	62220.000000	1.250000	
2	No	1	1.534154	32297.000000	2.166667	
6	Yes	0	1.592123	88512.500000	6.666667	
7	Yes	0	1.226408	31102.200000	5.583333	
9	Yes	0	1.720915	45702.000000	9.500000	
11	Yes	1	0.936553	34939.750000	4.666667	
13	Yes	0	0.814496	19148.800000	9.333333	
15	No	0	3.117344	76961.000000	2.583333	
16	No	0	0.214671	12848.500000	4.083333	
18	No	1	1.928263	26171.666667	9.916667	
19	Yes	0	1.305682	32229.200000	5.416667	
20	No	0	4.436136	66857.000000	8.000000	
21	No	0	0.688652	50310.500000	4.583333	
22	Yes	0	0.991322	26182.400000	4.000000	

	Age_Group	Risk_Score	HasFamilyDependents
0	Mature	-0.416497	1
1	Senior	-2.608657	0
2	Mature	-1.313565	1
6	Mid-age	-1.247417	0
7	Mature	-1.031702	0
9	Mid-age	-1.486209	0
11	Young	-1.295377	0
13	Mature	-1.079790	0
15	Mid-age	-3.499109	0
16	Young	-0.126436	0
18	Young	-1.722969	0
19	Mid-age	-1.253329	0
20	Senior	-4.581430	1

```
21    Mature   -0.575711                    1
22    Mature   -0.775440                    1
```

Married Borrowers with Dependents

In [46]:
```
married_dependents = df[(df['MaritalStatus'] == 'Married') &
                        (df['HasDependents'] == 'Yes')]
print(married_dependents.head(15))
```

```
        LoanID  Age    Income  LoanAmount  CreditScore  MonthsEmployed  \
21  A22KI1B6SE   47  146113.0    100621.0        419.0              55
22  1MUSHWD9TW   55  132058.0    130912.0        583.0              48
41  RT511ZZNF2   54   21487.0     84115.0        386.0              32
43  XG5WPXX0TY   24  105732.0     33013.0        400.0              58
44  2XUD7N4OJ1   38   73117.0    221628.0        639.0              84
46  Z7UFZIW3MK   56   24796.0     37657.0        498.0              88
49  LMQOF5PTTT   42  143244.0    240591.0        393.0              96
52  130U8TZ0HW   26   92153.0    214064.0        634.0              33
55  QHPWTGMTU8   37   37173.0    234059.0        455.0              20
61  Z3WJUIM1DZ   64  110606.0    137139.0        669.0              11
66  ZOPZLF57NR   67   77135.0    159839.0        529.0              59
68  ZY51VR44DK   21  142860.0    238718.0        318.0              35
70  MZRL2WMB52   23   17142.0    110469.0        802.0              56
75  5KCTUT4OSE   43  130625.0     17462.0        303.0              22
76  94DS7MC3PV   61   49191.0     93324.0        487.0              51

    NumCreditLines  InterestRate  LoanTerm  DTIRatio    Education  \
21               1          9.32        12      0.38   Bachelor'S
22               4          5.82        60      0.47  High School
41               2         20.89        60      0.78          Phd
43               3         15.64        36      0.48  High School
44               4         13.05        36      0.56   Bachelor'S
46               1          8.20        48      0.73  High School
49               3         23.15        36      0.16  High School
52               3          2.34        24      0.32          Phd
55               3         15.12        48      0.74          Phd
61               3         18.01        60      0.66     Master'S
66               4          4.14        36      0.41  High School
68               3         13.06        48      0.36  High School
70               3         12.23        12      0.71  High School
75               3         17.91        60      0.65     Master'S
76               2          2.65        12      0.25  High School

   EmploymentType MaritalStatus HasMortgage HasDependents LoanPurpose  \
21     Unemployed       Married         Yes           Yes    business
22     Unemployed       Married          No           Yes    business
41     Unemployed       Married         Yes           Yes    business
43       Fulltime       Married         Yes           Yes        auto
44       Parttime       Married          No           Yes        home
46       Parttime       Married         Yes           Yes        auto
```

49	Selfemployed	Married	No	Yes	business
52	Fulltime	Married	No	Yes	education
55	Fulltime	Married	No	Yes	other
61	Parttime	Married	No	Yes	home
66	Fulltime	Married	Yes	Yes	other
68	Parttime	Married	No	Yes	business
70	Fulltime	Married	Yes	Yes	auto
75	Selfemployed	Married	No	Yes	business
76	Unemployed	Married	No	Yes	home

	HasCoSigner	Default	Loan_to_Income	Credit_Utilization	Years_Employed	\
21	No	0	0.688652	50310.500000	4.583333	
22	Yes	0	0.991322	26182.400000	4.000000	
41	Yes	0	3.914693	28038.333333	2.666667	
43	Yes	0	0.312233	8253.250000	4.833333	
44	Yes	0	3.031142	44325.600000	7.000000	
46	No	0	1.518672	18828.500000	7.333333	
49	No	0	1.679589	60147.750000	8.000000	
52	No	0	2.322919	53516.000000	2.750000	
55	No	0	6.296479	58514.750000	1.666667	
61	No	0	1.239888	34284.750000	0.916667	
66	No	0	2.072198	31967.800000	4.916667	
68	Yes	0	1.670993	59679.500000	2.916667	
70	Yes	0	6.444347	27617.250000	4.666667	
75	No	0	0.133680	4365.500000	1.833333	
76	Yes	0	1.897176	31108.000000	4.250000	

	Age_Group	Risk_Score	HasFamilyDependents
21	Mature	-0.575711	1
22	Mature	-0.775440	1
41	Mature	-4.240575	1
43	Young	-0.321645	1
44	Mid-age	-2.839377	1
46	Mature	-1.662790	1
49	Mid-age	-1.377236	1
52	Young	-1.897037	1
55	Mid-age	-6.501185	1
61	Senior	-1.112829	1
66	Senior	-1.859845	1
68	Young	-1.656875	1
70	Young	-6.210818	1

```
75   Mid-age   -0.427210                    1
76    Senior   -1.574235                    1
```

High Interest Rate Loans

In [47]:
```
High_Interest_Rate_Loans = df[df['InterestRate'] > 15]
print(High_Interest_Rate_Loans.tail(15))
```

```
            LoanID  Age   Income  LoanAmount  CreditScore  MonthsEmployed  \
318835  I9X9Q5416F   43  82484.0    127506.0        574.0              85
318845  22IGDI7J5V   43  82484.0    127506.0        574.0              46
318859  J7CBWFLXFN   43  82484.0    127506.0        574.0              73
318890  PGVWWAD1VU   43  82484.0    127506.0        574.0             100
318928  BPJ2A0NEVA   43  82484.0    127506.0        574.0              43
318931  OK6MMILY5J   43  82484.0    127506.0        574.0              73
318986  TT14R1P1M7   43  82484.0    127506.0        574.0               5
319010  LBY7AT792Z   43  82484.0    127506.0        574.0              57
319017  SS4ACX50Q8   43  82484.0    127506.0        574.0              13
319025  F0NIB82WJL   43  82484.0    127506.0        574.0              36
319067  VJ7YALTZZN   43  82484.0    127506.0        574.0              46
319076  CPSCTZ5QGZ   43  82484.0    127506.0        574.0              58
319078  NVR7SNNIHQ   43  82484.0    127506.0        574.0              80
319086  4VPUOQDT79   43  82484.0    127506.0        574.0              22
319103  21Q7Q1LBT1   43  82484.0    127506.0        574.0              27

        NumCreditLines  InterestRate  LoanTerm  DTIRatio    Education  \
318835               3         19.77        12      0.57  High School
318845               4         19.79        12      0.48          Phd
318859               2         21.08        24      0.37  High School
318890               3         15.60        12      0.49   Bachelor'S
318928               4         17.21        48      0.18  High School
318931               3         16.98        48      0.33     Master'S
318986               2         20.77        48      0.30     Master'S
319010               4         23.13        12      0.34  High School
319017               4         18.05        24      0.63   Bachelor'S
319025               3         17.91        12      0.78   Bachelor'S
319067               4         19.93        12      0.89   Bachelor'S
319076               1         23.90        36      0.11          Phd
319078               4         21.18        48      0.38   Bachelor'S
319086               3         24.92        60      0.50          Phd
319103               1         23.16        12      0.78     Master'S

       EmploymentType MaritalStatus HasMortgage HasDependents LoanPurpose  \
318835       Parttime       Married          No           Yes   education
318845       Parttime      Divorced          No            No       other
318859   Selfemployed       Married          No            No        home
318890   Selfemployed       Married          No            No        auto
318928       Fulltime       Married          No            No        auto
318931       Fulltime       Married          No           Yes   education
```

318986	Parttime	Married	No	No	education
319010	Unemployed	Married	Yes	Yes	home
319017	Fulltime	Single	Yes	Yes	business
319025	Selfemployed	Married	No	Yes	home
319067	Fulltime	Divorced	Yes	Yes	home
319076	Unemployed	Single	Yes	No	other
319078	Selfemployed	Divorced	Yes	Yes	auto
319086	Parttime	Divorced	Yes	No	business
319103	Selfemployed	Divorced	No	No	home

	HasCoSigner	Default	Loan_to_Income	Credit_Utilization	\
318835	Yes	0	1.545827	31876.5	
318845	No	0	1.545827	25501.2	
318859	No	0	1.545827	42502.0	
318890	Yes	0	1.545827	31876.5	
318928	No	0	1.545827	25501.2	
318931	No	0	1.545827	31876.5	
318986	Yes	0	1.545827	42502.0	
319010	No	0	1.545827	25501.2	
319017	No	0	1.545827	25501.2	
319025	Yes	0	1.545827	31876.5	
319067	No	0	1.545827	25501.2	
319076	Yes	0	1.545827	63753.0	
319078	Yes	0	1.545827	25501.2	
319086	Yes	0	1.545827	31876.5	
319103	Yes	0	1.545827	63753.0	

	Years_Employed	Age_Group	Risk_Score	HasFamilyDependents
318835	7.083333	Mid-age	-1.440533	1
318845	3.833333	Mid-age	-1.350533	0
318859	6.083333	Mid-age	-1.240533	0
318890	8.333333	Mid-age	-1.360533	0
318928	3.583333	Mid-age	-1.050533	0
318931	6.083333	Mid-age	-1.200533	1
318986	0.416667	Mid-age	-1.170533	0
319010	4.750000	Mid-age	-1.210533	1
319017	1.083333	Mid-age	-1.500533	0
319025	3.000000	Mid-age	-1.650533	1
319067	3.833333	Mid-age	-1.760533	1
319076	4.833333	Mid-age	-0.980533	0
319078	6.666667	Mid-age	-1.250533	1

```
319086        1.833333   Mid-age   -1.370533                    0
319103        2.250000   Mid-age   -1.650533                    0
```

Senior Citizens

In [52]:
```
Senior_Citizens = df[df['Age'] > 60]
print(Senior_Citizens.head(20))
```

	LoanID	Age	Income	LoanAmount	CreditScore	MonthsEmployed	\
1	HPSK72WA7R	69.0	50432.0	124440.0	458.0	15	
20	7ASF0IHRIT	61.0	30142.0	133714.0	429.0	96	
28	BJNLQ0H95H	61.0	62519.0	29676.0	462.0	16	
30	GAA8OQN796	66.0	39568.0	58945.0	604.0	37	
36	8NTWNU4HTY	64.0	102463.0	218433.0	506.0	24	
37	ASYFXCP452	68.0	85409.0	44772.0	540.0	105	
38	4857M8R0YI	61.0	26470.0	19818.0	695.0	47	
39	5FENBP2UV8	69.0	87295.0	16281.0	707.0	94	
42	KAC7P2RE1X	68.0	111716.0	215851.0	747.0	99	
51	N3KDA4UM9K	67.0	100949.0	145906.0	439.0	36	
57	DSOB0M5AQ4	64.0	49565.0	245711.0	655.0	76	
59	0VEK3DDM69	61.0	49113.0	222046.0	771.0	47	
61	Z3WJUIM1DZ	64.0	110606.0	137139.0	669.0	11	
66	ZOPZLF57NR	67.0	77135.0	159839.0	529.0	59	
76	94DS7MC3PV	61.0	49191.0	93324.0	487.0	51	
82	SQ3MA71EVN	65.0	80770.0	217916.0	574.0	9	
90	I7Z9O3RG38	62.0	138569.0	248333.0	453.0	101	
92	CL2BS2EEJE	64.0	93741.0	232727.0	747.0	44	
96	D51F6APQ5Y	62.0	25274.0	134289.0	701.0	91	
100	XL0LIO7XMD	62.0	47383.0	13109.0	553.0	73	

	NumCreditLines	InterestRate	LoanTerm	DTIRatio	Education	\
1	1	4.81	60	0.68	Master'S	
20	1	15.58	12	0.65	Phd	
28	1	23.91	48	0.12	Bachelor'S	
30	4	6.67	12	0.10	High School	
36	2	9.23	60	0.86	Master'S	
37	1	2.96	36	0.17	Master'S	
38	2	20.00	36	0.69	Phd	
39	1	13.82	60	0.75	Bachelor'S	
42	3	18.84	60	0.40	High School	
51	2	3.74	48	0.58	Bachelor'S	
57	4	23.41	24	0.37	High School	
59	3	22.80	12	0.24	High School	
61	3	18.01	60	0.66	Master'S	
66	4	4.14	36	0.41	High School	
76	2	2.65	12	0.25	High School	
82	3	20.97	36	0.49	High School	
90	2	14.10	24	0.44	High School	
92	1	24.29	24	0.62	Master'S	

```
96                1         13.01        24      0.57     Master'S
100               3         19.57        12      0.42   Bachelor'S

    EmploymentType MaritalStatus HasMortgage HasDependents LoanPurpose  \
1        Full-time       Married          No            No       other
20       Part-time      Divorced          No           Yes    business
28      Unemployed      Divorced         Yes            No        home
30      Unemployed      Divorced         Yes           Yes        auto
36      Unemployed       Married          No            No        auto
37      Unemployed      Divorced          No            No        auto
38      Unemployed      Divorced         Yes           Yes        auto
39       Part-time        Single          No            No       other
42      Unemployed      Divorced          No           Yes        home
51      Unemployed       Married          No            No        auto
57      Unemployed        Single          No           Yes        auto
59      Unemployed      Divorced          No           Yes       other
61       Part-time       Married          No           Yes        home
66       Full-time       Married         Yes           Yes       other
76      Unemployed       Married          No           Yes        home
82       Part-time        Single         Yes            No        home
90       Part-time       Married         Yes           Yes       other
92       Full-time      Divorced         Yes           Yes       other
96      Unemployed       Married          No            No    business
100     Unemployed       Married         Yes            No    business

    HasCoSigner  Default
1           Yes        0
20           No        0
28          Yes        0
30          Yes        0
36          Yes        0
37           No        0
38          Yes        0
39           No        0
42           No        0
51          Yes        0
57          Yes        0
59           No        0
61           No        0
66           No        0
76          Yes        0
```

```
82           No        0
90          Yes        0
92           No        0
96          Yes        0
100          No        0
```

Months Employed less than 1 year

In [48]:
```
Less_Than_Year=df[df['MonthsEmployed'] < 12]
print(Less_Than_Year.head(5))
```

```
        LoanID  Age    Income  LoanAmount  CreditScore  MonthsEmployed  \
3   V2KKSFM3UN   32   31713.0     44799.0        743.0               0
4   EY08JDHTZP   60   20437.0      9139.0        633.0               8
32  KD97QJJFD8   59  102292.0     55337.0        840.0               6
47  RSP1YD80Z7   35   95963.0     77552.0        560.0               8
53  E9NX4IRVSU   43   70113.0    215064.0        657.0               0

    NumCreditLines  InterestRate  LoanTerm  DTIRatio    Education  \
3                3          7.07        24      0.23  High School
4                4          6.51        48      0.73   Bachelor'S
32               1         16.11        60      0.44     Master'S
47               2          6.63        24      0.86     Master'S
53               2          6.00        24      0.76   Bachelor'S

   EmploymentType MaritalStatus HasMortgage HasDependents LoanPurpose  \
3        Fulltime       Married          No            No    business
4      Unemployed      Divorced          No           Yes        auto
32     Unemployed       Married         Yes            No        auto
47   Selfemployed      Divorced         Yes           Yes        home
53   Selfemployed        Single          No           Yes        home

   HasCoSigner  Default  Loan_to_Income  Credit_Utilization  Years_Employed  \
3           No        0        1.412638        11199.750000        0.000000
4           No        0        0.447179         1827.800000        0.666667
32          No        0        0.540971        27668.500000        0.500000
47          No        1        0.808145        25850.666667        0.666667
53         Yes        0        3.067391        71688.000000        0.000000

   Age_Group  Risk_Score  HasFamilyDependents
3    Mid-age   -0.768521                    0
4     Mature   -0.432473                    1
32    Mature    0.007264                    0
47   Mid-age   -1.009321                    1
53   Mid-age   -3.054450                    0
```

High DTI Ratio

In [49]:
```
High_DTI_Ratio = df[df['DTIRatio'] > 0.5] # (>0.5-> risky)
print(High_DTI_Ratio.tail(10))
```

	LoanID	Age	Income	LoanAmount	CreditScore	MonthsEmployed	\
319024	6BEKC0R79X	43	82484.0	127506.0	574.0	13	
319025	F0NIB82WJL	43	82484.0	127506.0	574.0	36	
319054	12IT24XQX5	43	82484.0	127506.0	574.0	111	
319066	0ZYOV6QRT8	43	82484.0	127506.0	574.0	45	
319067	VJ7YALTZZN	43	82484.0	127506.0	574.0	46	
319088	SFU0VSRSIU	43	82484.0	127506.0	574.0	57	
319094	Q5QSJCH8PO	43	82484.0	127506.0	574.0	28	
319103	21Q7Q1LBT1	43	82484.0	127506.0	574.0	27	
319130	LQT2S0YRTT	43	82484.0	127506.0	574.0	82	
319174	X8W1CFI5N4	43	82484.0	127506.0	574.0	48	

	NumCreditLines	InterestRate	LoanTerm	DTIRatio	Education	\
319024	4	10.74	60	0.51	High School	
319025	3	17.91	12	0.78	Bachelor'S	
319054	2	10.32	48	0.68	High School	
319066	2	11.99	12	0.72	High School	
319067	4	19.93	12	0.89	Bachelor'S	
319088	4	10.34	60	0.82	High School	
319094	4	7.08	12	0.59	Phd	
319103	1	23.16	12	0.78	Master'S	
319130	2	6.22	36	0.75	Master'S	
319174	4	14.35	48	0.71	Master'S	

	EmploymentType	MaritalStatus	HasMortgage	HasDependents	LoanPurpose	\
319024	Parttime	Divorced	No	Yes	education	
319025	Selfemployed	Married	No	Yes	home	
319054	Selfemployed	Married	Yes	Yes	education	
319066	Selfemployed	Divorced	Yes	Yes	business	
319067	Fulltime	Divorced	Yes	Yes	home	
319088	Fulltime	Single	Yes	No	education	
319094	Unemployed	Single	No	Yes	home	
319103	Selfemployed	Divorced	No	No	home	
319130	Unemployed	Divorced	No	No	auto	
319174	Selfemployed	Single	No	No	business	

	HasCoSigner	Default	Loan_to_Income	Credit_Utilization	\
319024	No	0	1.545827	25501.2	
319025	Yes	0	1.545827	31876.5	
319054	No	0	1.545827	42502.0	
319066	No	0	1.545827	42502.0	

```
319067          No        0        1.545827             25501.2
319088         Yes        0        1.545827             25501.2
319094          No        0        1.545827             25501.2
319103         Yes        0        1.545827             63753.0
319130         Yes        0        1.545827             42502.0
319174          No        0        1.545827             25501.2

        Years_Employed Age_Group  Risk_Score  HasFamilyDependents
319024        1.083333   Mid-age   -1.380533                    1
319025        3.000000   Mid-age   -1.650533                    1
319054        9.250000   Mid-age   -1.550533                    1
319066        3.750000   Mid-age   -1.590533                    1
319067        3.833333   Mid-age   -1.760533                    1
319088        4.750000   Mid-age   -1.690533                    0
319094        2.333333   Mid-age   -1.460533                    0
319103        2.250000   Mid-age   -1.650533                    0
319130        6.833333   Mid-age   -1.620533                    0
319174        4.000000   Mid-age   -1.580533                    0
```

➡️ Aggregating Data

Loan Purposes By Count

In [50]:
```
#df['LoanPurpose'] → selects the LoanPurpose column.
#.value_counts() → counts how many times each loan purpose appears.
#.head() → keeps only the top most frequent.
loan_purposes = df['LoanPurpose'].value_counts().head()
print("loan_Purposes by count:")
print(loan_purposes)
```

```
loan_Purposes by count:
LoanPurpose
home         52612
business     52543
education    52233
other        52211
auto         52132
Name: count, dtype: int64
```

Average Loan Amount by Education

In [51]:
```
#df.groupby("Education") → groups data by Education level.
#["LoanAmount"].mean() → calculates average LoanAmount for each group.
#.sort_values(ascending=False) → sorts from highest to lowest.
#.head() → takes the most frequent education levels.
avg_loan_by_edu = df.groupby("Education")["LoanAmount"].mean().sort_values(ascending=False).head()
print("\nAverage Loan Amount by Education:")
print(avg_loan_by_edu)
```

```
Average Loan Amount by Education:
Education
Phd            127816.239760
Master'S       127706.052897
High School    127396.397044
Bachelor'S     127393.612565
Name: LoanAmount, dtype: float64
```

Default Rate by Marital Status

In [52]:
```
#Groups borrowers by MaritalStatus.
#["Default"].mean() → gives the proportion of defaults (since Default is 0 or 1).
#Sorted highest to lowest.
default_by_marital = df.groupby("MaritalStatus")["Default"].mean().sort_values(ascending=False)
print("\nDefault Rate by Marital Status:")
print(default_by_marital)
```

```
Default Rate by Marital Status:
MaritalStatus
Divorced    0.125220
Single      0.119128
Married     0.103956
Name: Default, dtype: float64
```

Average Credit Score by Loan Purpose

In [53]:
```
#Groups by LoanPurpose.
#Computes the average credit score for each loan type.
#Sorted descending,most frequent
avg_credit_by_purpose = df.groupby("LoanPurpose")["CreditScore"].mean().sort_values(ascending=False).head(10)
print("\nAverage Credit Score by Loan Purpose:")
print(avg_credit_by_purpose)
```

```
Average Credit Score by Loan Purpose:
LoanPurpose
home         575.034270
auto         574.561690
other        574.397732
business     574.060636
education    573.231348
Name: CreditScore, dtype: float64
```

Employment Types by Count

In [54]:
```
#Shows which job types occur most often.
employment_types = df['EmploymentType'].value_counts().head()
print("\nEmployment Types by count:")
print(employment_types)
```

```
Employment Types by count:
EmploymentType
Parttime        65733
Selfemployed    65381
Unemployed      65378
Fulltime        65239
Name: count, dtype: int64
```

---

3. Exploratory Data Analysis (EDA)

➡️ Conduct descriptive and exploratory analysis to uncover pattern and trends :

➡️ Univariate,Bivariate and Multivariate Analysis

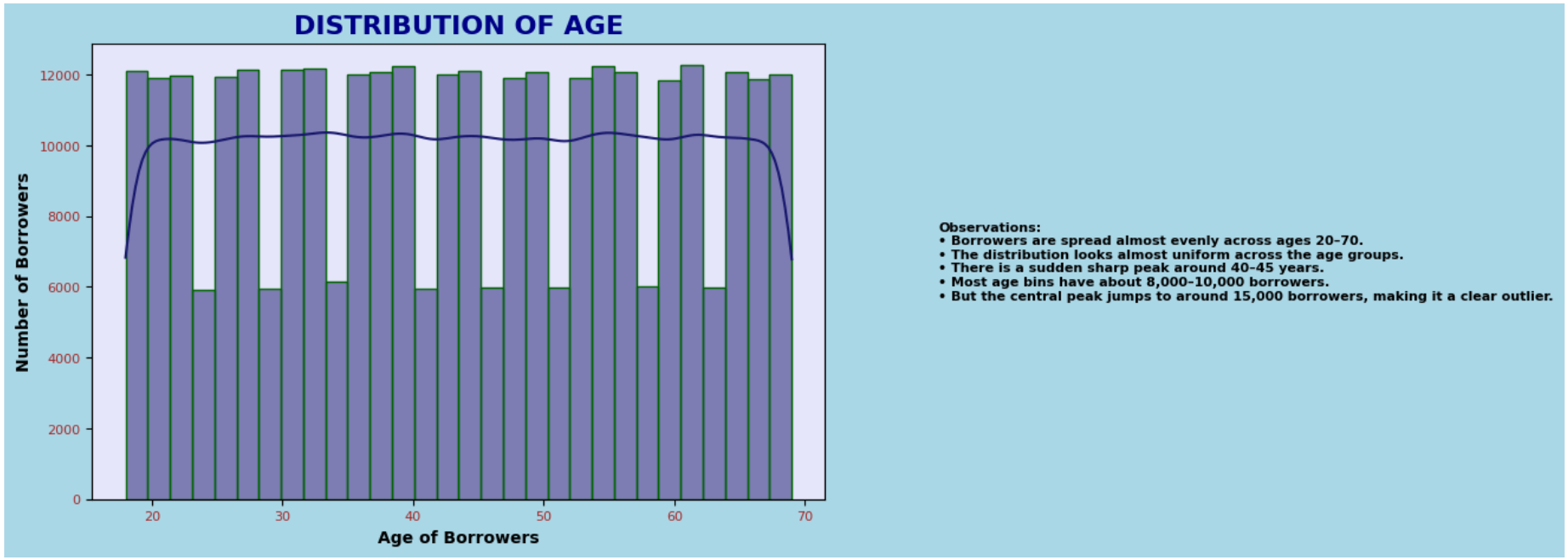
In [ ]:
```
#Univariate (Single Variable Analysis)-Understand distribution,characteristics of one column at a time. Helps detect outliers,
#Bivariate  (Two Variable Analysis)-Explore the relationship between two variables at a time. Shows correlations and patterns
#Multivariate (Three or more variables together)-Analyze how multiple factors interact together to affect the target variable
```

✅ Univariate Analysis(Single Variable Analysis)

Age

In [2]:
```
# Distribution of Borrower age-> Are most borrowers middle aged or not?
import matplotlib.pyplot as plt
import seaborn as sns

#The bars show how many borrowers fall into different age groups.

# Histogram
plt.figure(figsize=(8,5),facecolor="lightblue")  # background outside plot

sns.histplot(df['Age'], kde=True, bins=30, color="midnightblue",edgecolor="darkgreen") #KDE->Kernel Density Estimation:if true

# Title
plt.title("DISTRIBUTION OF AGE", fontsize=16,weight="bold", color="darkblue")

# X and Y axis labels
plt.xlabel("Age of Borrowers", fontsize=10,weight="bold", color="black")
plt.ylabel("Number of Borrowers", fontsize=10,weight="bold", color="black")

# Change background inside plot
plt.gca().set_facecolor("lavender")  #gca(Get Current Axes)->the current plot area inside the figure.
                                     #.set_facecolor- changes the background colour of the plot area.
#Side note inside the figure
plt.figtext(1.02,0.5,
            "Observations:\n"
            "• Borrowers are spread almost evenly across ages 20–70.\n"
            "• The distribution looks almost uniform across the age groups.\n"
            "• There is a sudden sharp peak around 40–45 years.\n"
            "• Most age bins have about 8,000–10,000 borrowers.\n"
            "• But the central peak jumps to around 15,000 borrowers, making it a clear outlier.\n",

         ha="left", va="center", fontsize=8,weight ="bold", color="black")




# Change tick labels size and color
plt.xticks(color="brown", fontsize=8)
plt.yticks(color="brown", fontsize=8)
```

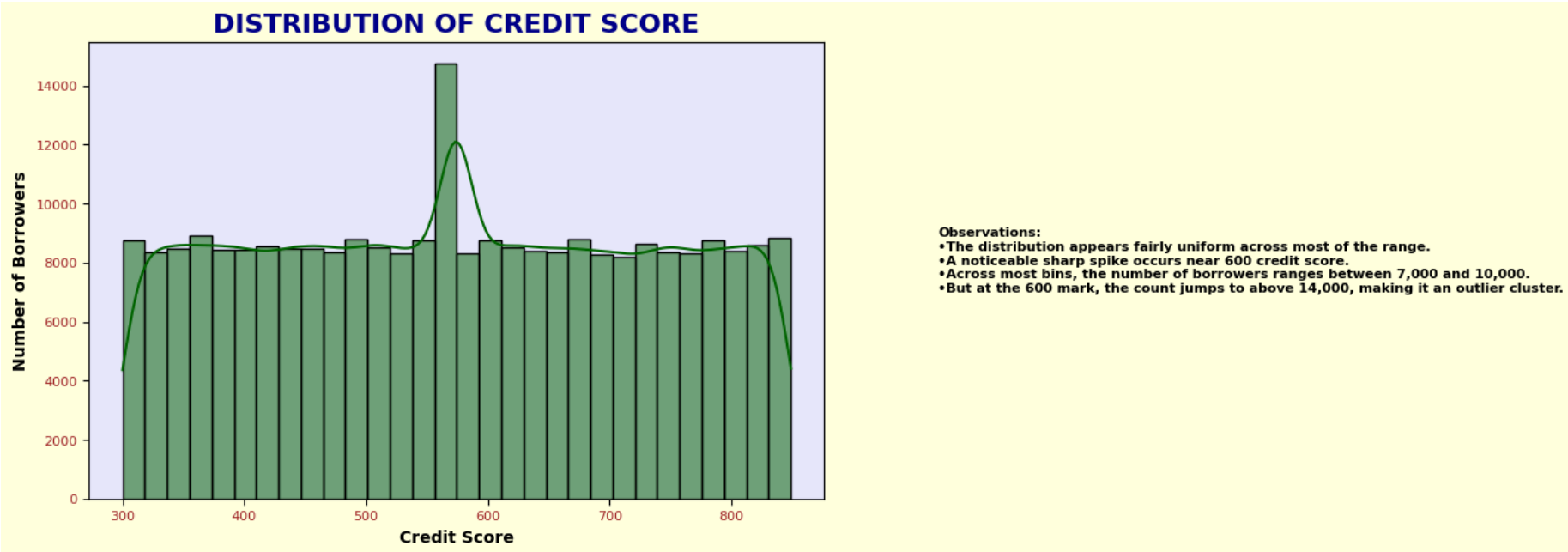
```
plt.show()
```



CreditScore

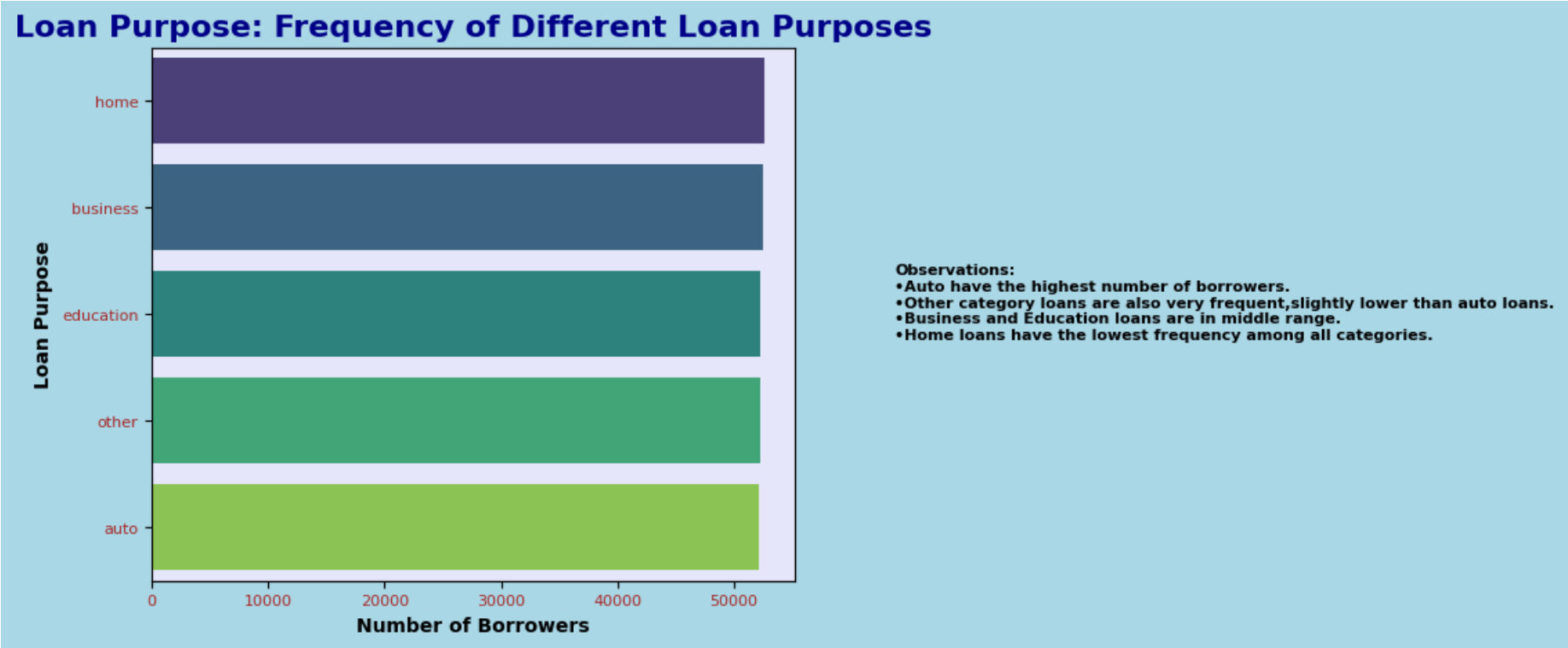
In [74]:
```
#Check if scores are normally distributed or skewed.


# Histogram
plt.figure(figsize=(8,5), facecolor="lightyellow")  # background outside plot

# Histogram for Credit Score
sns.histplot(df['CreditScore'], kde=True, bins=30, color="darkgreen")

# Title
plt.title("DISTRIBUTION OF CREDIT SCORE", fontsize=16,weight="bold", color="darkblue")

# X and Y axis labels
plt.xlabel("Credit Score", fontsize=10,weight="bold", color="black")
plt.ylabel("Number of Borrowers", fontsize=10,weight="bold", color="black")
```

```
# Change background inside plot
plt.gca().set_facecolor("lavender")


#Side note inside the figure
plt.figtext(1.02,0.5,
            "Observations:\n"
            "•The distribution appears fairly uniform across most of the range. \n"
            "•A noticeable sharp spike occurs near 600 credit score. \n"
            "•Across most bins, the number of borrowers ranges between 7,000 and 10,000.\n"
                "•But at the 600 mark, the count jumps to above 14,000, making it an outlier cluster.\n",
         ha="left", va="center", fontsize=8,weight="bold", color="black")



# Change tick labels size and color
plt.xticks(color="brown", fontsize=8)
plt.yticks(color="brown", fontsize=8)

plt.show()
```



Loan Purpose

In [76]:
```
# Frequency of different loan purposes.
plt.figure(figsize=(6,5), facecolor="lightblue")  # Background outside plot

# Bar chart for Loan Purpose
sns.countplot(
    y='LoanPurpose', #bars arranged horizontally
    data=df,
    order=df['LoanPurpose'].value_counts().index, #Bars sorted by frequency(most frequent on top)
    palette="viridis" #Uses the viridis color theme.
)

# Title
plt.title("Loan Purpose: Frequency of Different Loan Purposes", fontsize=16,weight="bold", color="darkblue")

# X and Y axis labels
plt.xlabel("Number of Borrowers", fontsize=10,weight="bold", color="black")
plt.ylabel("Loan Purpose", fontsize=10,weight="bold", color="black")

# Change background inside plot
plt.gca().set_facecolor("lavender")

#Side note inside the figure
plt.figtext(1.02,0.5,
            "Observations:\n"
            "•Auto have the highest number of borrowers. \n"
            "•Other category loans are also very frequent,slightly lower than auto loans.\n"
            "•Business and Education loans are in middle range. \n"
            "•Home loans have the lowest frequency among all categories.\n",
         ha="left", va="center", fontsize=8,weight="bold", color="black")


# Change tick labels size and color
plt.xticks(color="brown", fontsize=8)
plt.yticks(color="brown", fontsize=8)

plt.show()
```

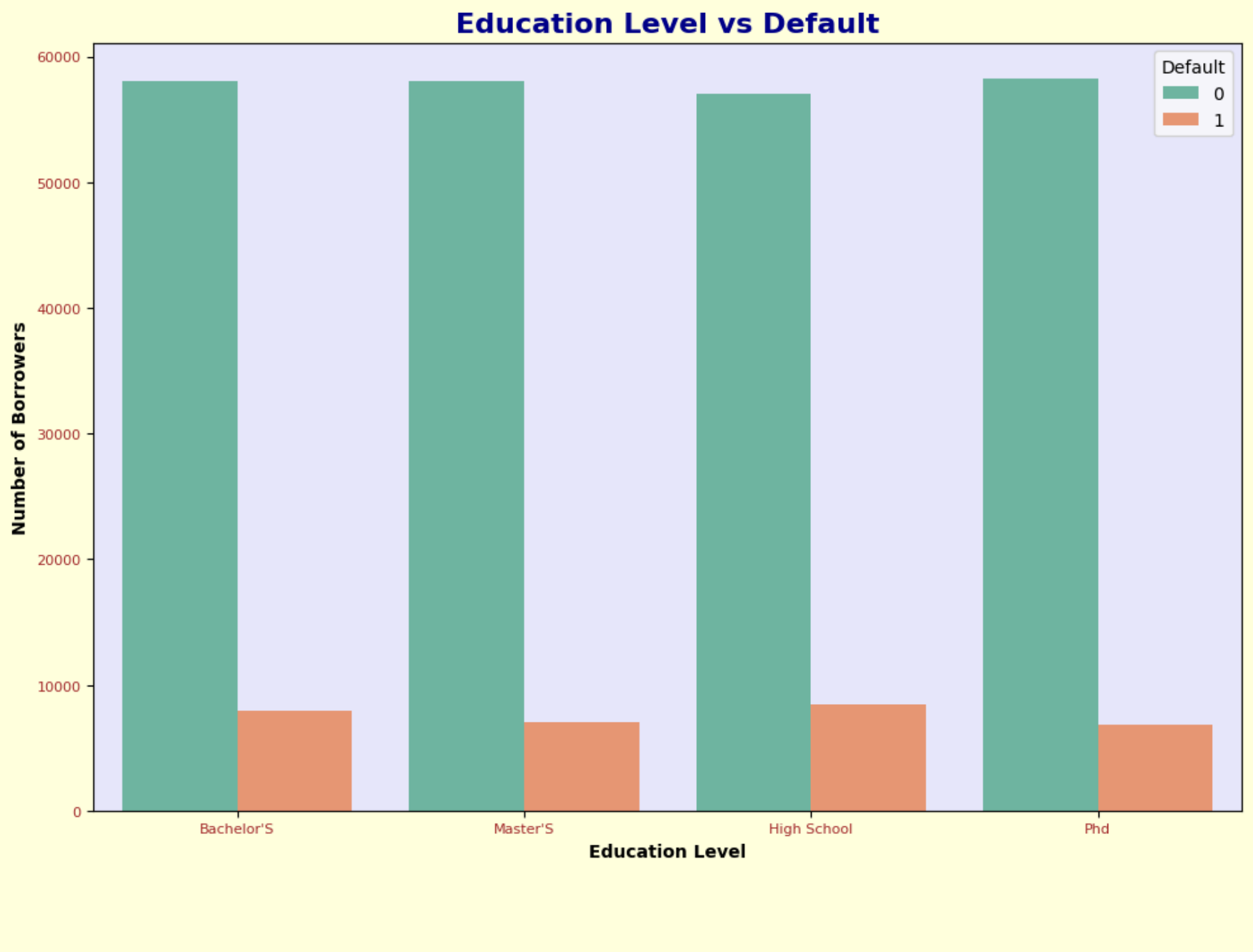


✅ Bivariate Analysis(Two Variable Analysis)

Education vs Default

In [78]:
```
#it’s comparing how many borrowers in each education group defaulted.
plt.figure(figsize=(10,7), facecolor="lightyellow")

sns.countplot(
    x="Education",
    hue="Default",
    data=df,
    palette="Set2"
)

plt.title("Education Level vs Default",
          fontsize=16, weight="bold", color="darkblue")
```

```
plt.xlabel("Education Level", fontsize=10, weight="bold", color="black")
plt.ylabel("Number of Borrowers", fontsize=10, weight="bold", color="black")

plt.gca().set_facecolor("lavender")


plt.figtext(0.1, -0.15,
         "Observations:\n"
          "•Majority of borrowers did not default (Default = 0) across all education levels. \n"
          "•Bachelor's degree show a relatively higher number of defaults compared to other education categories (high school,
          "•Master’s borrowers form slightly fewer borrowers than bachelor's,with defaults but a lit lower.\n"
          "•High school borrowers are comparable to bachelor's and master's,but default numbers are slightly higher than maste
          "•phD holders have lowest among all.\n",
          ha="left", va="center",fontsize=8,weight="bold", color="black")


plt.xticks(color="brown", fontsize=8)
plt.yticks(color="brown", fontsize=8)

plt.tight_layout()
plt.show()
```

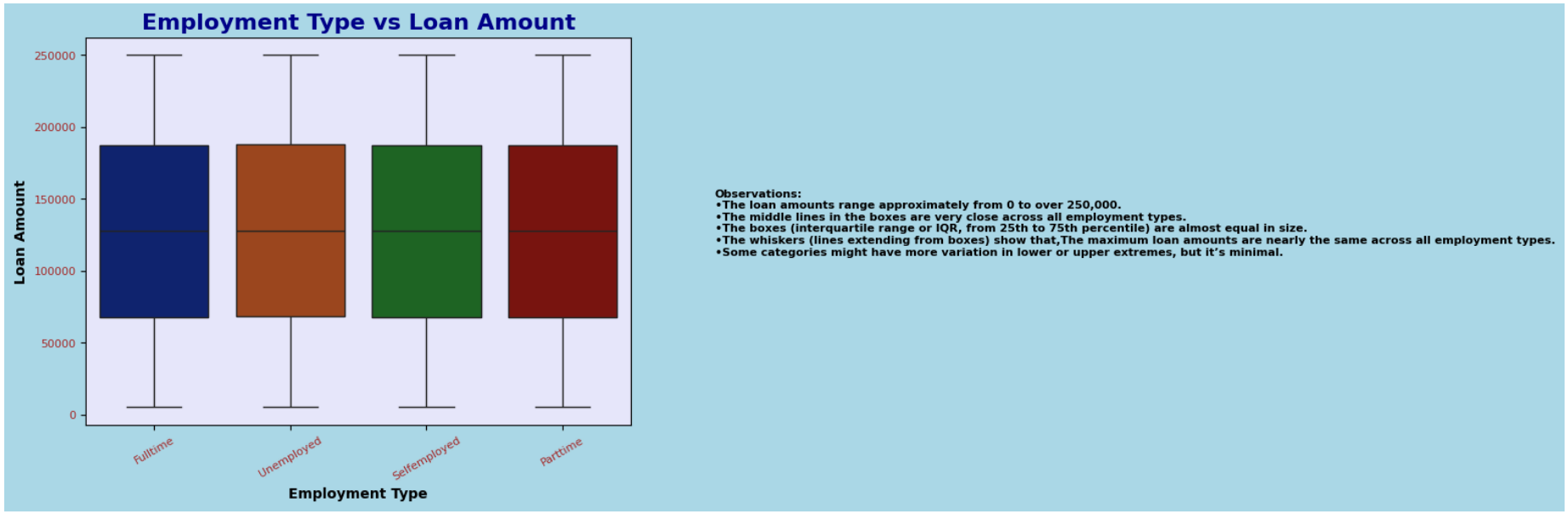

Education Level vs Default
Default
0
1
60000
50000
40000
30000
20000
10000
0
Number of Borrowers
Bachelor'S
Master'S
High School
Phd
Education Level

Observations:
•Majority of borrowers did not default (Default = 0) across all education levels.
•Bachelor's degree show a relatively higher number of defaults compared to other education categories (high school, Master's, PhD).
•Master's borrowers form slightly fewer borrowers than bachelor's,with defaults but a lit lower.
•High school borrowers are comparable to bachelor's and master's,but default numbers are slightly higher than master's.
•phD holders have lowest among all.

EmploymentType vs LoanAmount

In [79]:
```
# It checks about borrower's employment type affect the loan amount they take
#Boxplot is best when one variable is categorical (EmploymentType) and another is numeric (LoanAmount).
plt.figure(figsize=(7,5), facecolor="lightblue")

# Create boxplot (EmploymentType vs LoanAmount)
sns.boxplot(
    x="EmploymentType",
    y="LoanAmount",
    data=df,
    palette="dark"   )

plt.title("Employment Type vs Loan Amount", fontsize=16, weight="bold", color="darkblue")

# X and Y axis labels
plt.xlabel("Employment Type", fontsize=10, weight="bold")
plt.ylabel("Loan Amount", fontsize=10, weight="bold")

plt.gca().set_facecolor("lavender")

plt.xticks(rotation=30,color="brown",fontsize=8)
plt.yticks(color="brown",fontsize=8)

# observations noted at the side
plt.figtext(1.02,0.5,
            "Observations:\n"
            "•The loan amounts range approximately from 0 to over 250,000. \n"
            "•The middle lines in the boxes are very close across all employment types. \n"
            "•The boxes (interquartile range or IQR, from 25th to 75th percentile) are almost equal in size. \n"
            "•The whiskers (lines extending from boxes) show that,The maximum loan amounts are nearly the same across all empl
            "•Some categories might have more variation in lower or upper extremes, but it's minimal.\n",
         ha="left", va="center", fontsize=8, weight="bold", color="black")
```

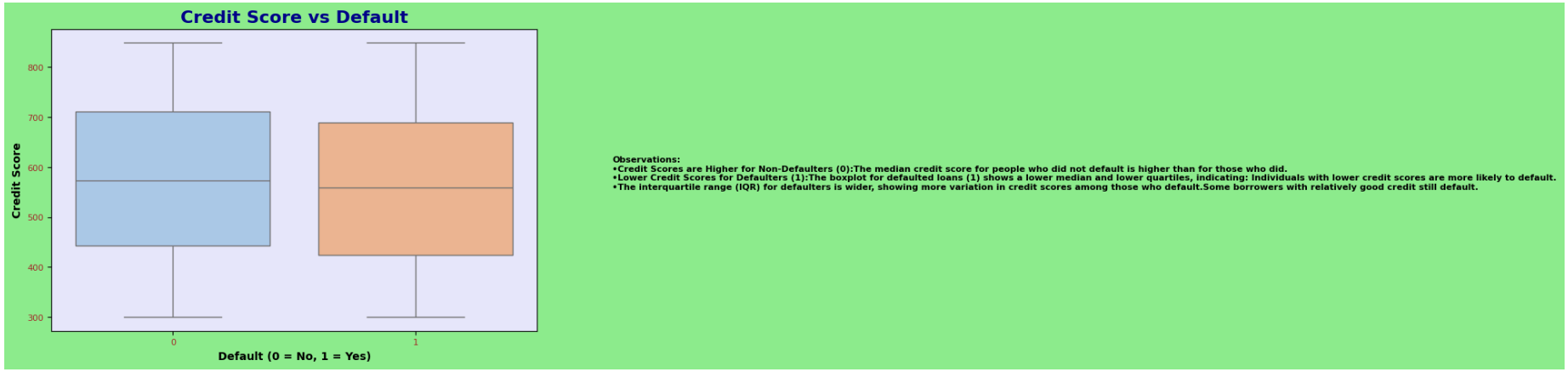
```
plt.show()
```



CreditScore vs Default

In [80]:
```
#This is used to find out how reliable credit score is for predicting loan repayment risk.
plt.figure(figsize=(8,5), facecolor="lightgreen")

# Boxplot: comparing Credit Score distribution across Default groups
sns.boxplot(
    x="Default",                # Default (0 = No, 1 = Yes) is on X-axis
    y="CreditScore",
    data=df,
    palette="pastel"
)

plt.title("Credit Score vs Default", fontsize=16, weight="bold", color="darkblue")
plt.xlabel("Default (0 = No, 1 = Yes)", fontsize=10, weight="bold")
plt.ylabel("Credit Score", fontsize=10, weight="bold")

# observations noted at the side
plt.figtext(1.02,0.5,
```

```
            "Observations:\n"
            "•Credit Scores are Higher for Non-Defaulters (0):The median credit score for people who did not default is higher
            "•Lower Credit Scores for Defaulters (1):The boxplot for defaulted loans (1) shows a lower median and lower quarti
            "•The interquartile range (IQR) for defaulters is wider, showing more variation in credit scores among those who d
            ha="left", va="center", fontsize=8, weight="bold", color="black")
plt.xticks(color="brown",fontsize=8)
plt.yticks(color="brown",fontsize=8)

plt.gca().set_facecolor("lavender")
plt.show()
```

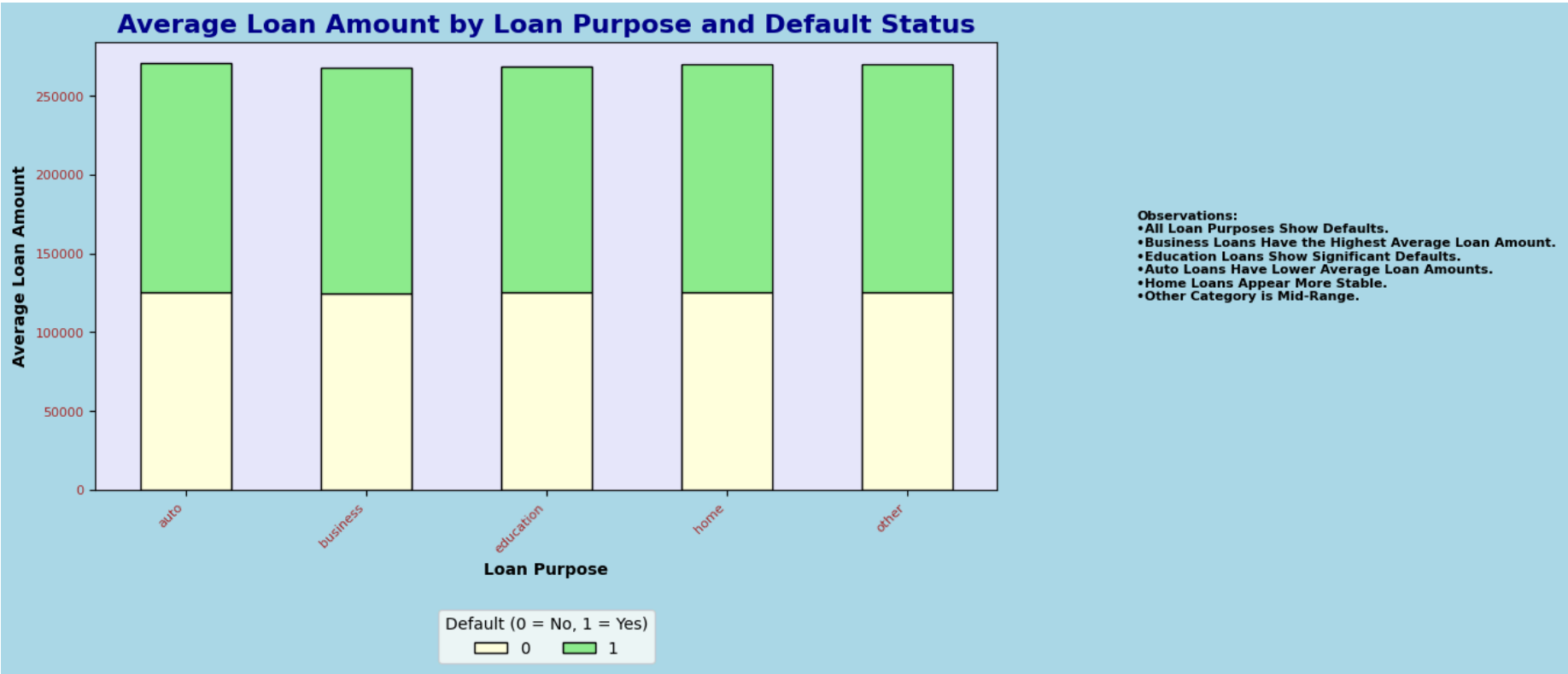


✅ Multivariate Analysis(Three or more Variable Analysis)

LoanPurpose vs LoanAmount vs Default

In [82]:
```
# For each type of loan, what is the average loan amount, and how is it distributed between defaulters and non-defaulters.
# Group the data: LoanPurpose vs LoanAmount grouped by Default
grouped = df.groupby(["LoanPurpose", "Default"])["LoanAmount"].mean().unstack()
# df.groupby(["LoanPurpose", "Default"]) → groups the dataset first by LoanPurpose, then by Default status.
# ["LoanAmount"].mean() → calculates the average LoanAmount for each group
# .unstack() → reshapes the table so that Default values (0 = No, 1 = Yes) become separate columns.
## This gives us a table where rows = LoanPurpose and columns = Default status, values = mean loan amount.
plt.figure(figsize=(10,5), facecolor="lightblue")

# Stacked bar chart
grouped.plot(        # plots the grouped data.
    kind="bar",      # makes a bar chart.
```

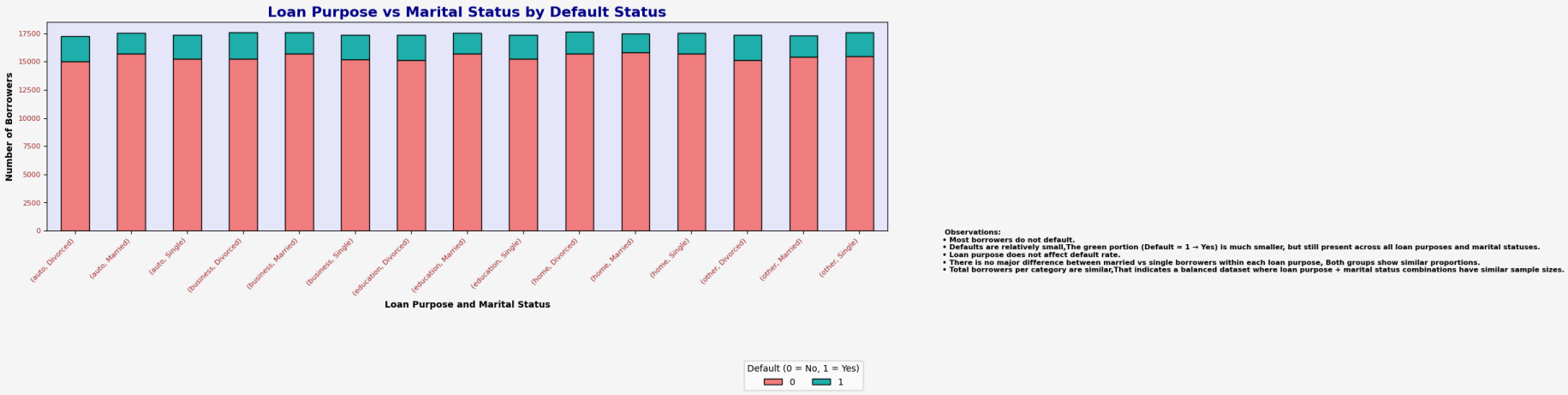
```
    stacked=True,    # stacks Default=0 and Default=1 values on top of each other inside each LoanPurpose bar
    color=["lightyellow", "lightgreen"],
    edgecolor="black",
    ax=plt.gca()
)


plt.title("Average Loan Amount by Loan Purpose and Default Status", fontsize=16, weight="bold", color="darkblue")
plt.xlabel("Loan Purpose", fontsize=10, weight="bold")
plt.ylabel("Average Loan Amount", fontsize=10, weight="bold")

plt.legend(    # legend explains colors, line styles
    title="Default (0 = No, 1 = Yes)", # sets a title above the legend
    loc="upper center", # legend box location
    bbox_to_anchor=(0.5, -0.25), # adjustments done in this legend box, centered hporizontally and pushed slightly below the a
    ncol=2) # legend items in 2 columns without stacking.


# observations noted at the side
plt.figtext(1.02,0.5,
            "Observations:\n"
            "•All Loan Purposes Show Defaults.\n"
            "•Business Loans Have the Highest Average Loan Amount.\n"
            "•Education Loans Show Significant Defaults.\n"
            "•Auto Loans Have Lower Average Loan Amounts.\n"
            "•Home Loans Appear More Stable.\n"
            "•Other Category is Mid-Range.\n",
             ha="left", va="center", fontsize=8, weight="bold", color="black")

plt.gca().set_facecolor("lavender")


plt.xticks(color="brown",fontsize=8,rotation=45, ha="right")
plt.yticks(color="brown",fontsize=8)

plt.show()
```



LoanPurpose vs MaritalStatus vs Default

In [86]:
```
# Helps  to see which loan purposes and marital statuses have higher default rates.
# Group the data: LoanPurpose vs MaritalStatus grouped by Default.
grouped = df.groupby(["LoanPurpose", "MaritalStatus", "Default"]).size().unstack()

plt.figure(figsize=(14,8), facecolor="whitesmoke")

# stacked bar chart
grouped.plot(
    kind="bar",
    stacked=True,
    color=["lightcoral", "lightseagreen"],
    edgecolor="black",
    ax=plt.gca()
)
```

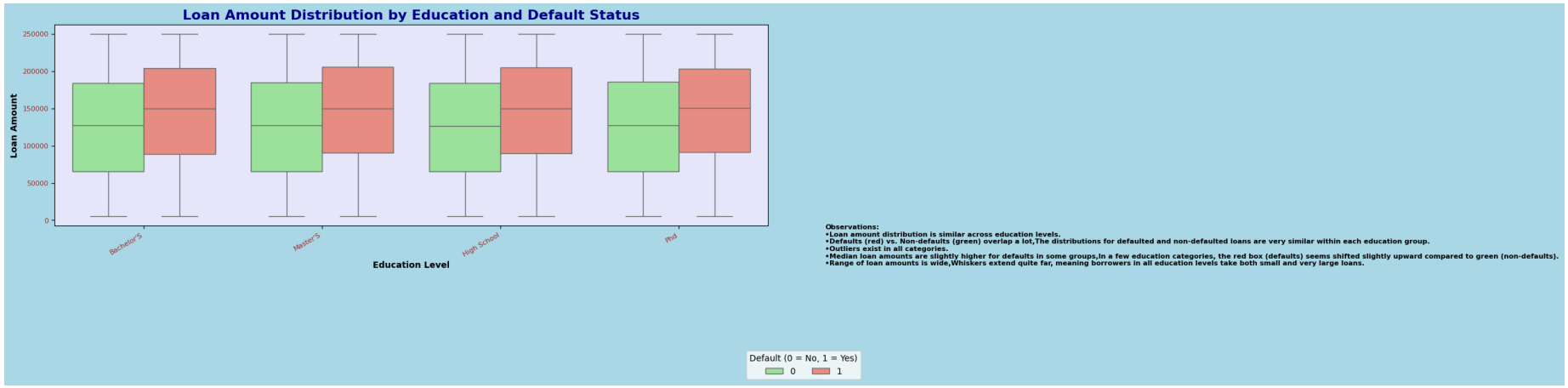
```
plt.title("Loan Purpose vs Marital Status by Default Status",
          fontsize=16, weight="bold", color="darkblue")

plt.xlabel("Loan Purpose and Marital Status", fontsize=10, weight="bold")

plt.ylabel("Number of Borrowers", fontsize=10, weight="bold")

plt.legend(title="Default (0 = No, 1 = Yes)", fontsize=8)

plt.gca().set_facecolor("lavender")

plt.xticks(color="brown",fontsize=8,rotation=45, ha="right")
plt.yticks(color="brown",fontsize=8)

# observations noted at the side
plt.figtext(1.05,0.5,
            " Observations:\n"
            "• Most borrowers do not default.\n"
            "• Defaults are relatively small,The green portion (Default = 1 → Yes) is much smaller, but still present across a
            "• Loan purpose does not affect default rate.\n"
            "• There is no major difference between married vs single borrowers within each loan purpose, Both groups show sim
            "• Total borrowers per category are similar,That indicates a balanced dataset where loan purpose + marital status
            ha="left", va="center", fontsize=8, weight="bold", color="black")

plt.legend(
    title="Default (0 = No, 1 = Yes)",
    loc="upper center",
    bbox_to_anchor=(0.9, -0.60),
    ncol=2)

plt.tight_layout()
plt.show()
```



In [ ]:
```
<h6 style="color:Blue;">LoanAmount vs Education vs Default</h6>
```

In [88]:
```
# Shows whether education level impacts loan amounts and default.
plt.figure(figsize=(14,8), facecolor="lightblue")

sns.boxplot(
    data=df,
    x="Education",
    y="LoanAmount",
    hue="Default",       # Split by Default status (0 = No, 1 = Yes)
    palette={0: "lightgreen", 1: "salmon"}  # Colors for Default groups
)
# → Boxplot shows distribution of LoanAmount within each Education level
# → Hue splits each box into two groups (Default vs Non-default)
# → Green = Non-default, Red = Default

plt.title("Loan Amount Distribution by Education and Default Status",
          fontsize=16, weight="bold", color="darkblue")

plt.xlabel("Education Level", fontsize=10, weight="bold")

plt.ylabel("Loan Amount", fontsize=10, weight="bold")

plt.xticks(color="brown",fontsize=8,rotation=30, ha="right")
plt.yticks(color="brown",fontsize=8)
```

```
plt.gca().set_facecolor("lavender")

# observations noted at the side
plt.figtext(0.95,0.5,
            "Observations:\n"
            "•Loan amount distribution is similar across education levels.\n"
            "•Defaults (red) vs. Non-defaults (green) overlap a lot,The distributions for defaulted and non-defaulted loans ar
            "•Outliers exist in all categories.\n"
            "•Median loan amounts are slightly higher for defaults in some groups,In a few education categories, the red box (
            "•Range of loan amounts is wide,Whiskers extend quite far, meaning borrowers in all education levels take both sma
            ha="left", va="center", fontsize=8, weight="bold", color="black")

#plt.legend(title="Default (0 = No, 1 = Yes)", fontsize=9)
plt.legend(
    title="Default (0 = No, 1 = Yes)",
    loc="upper center",
    bbox_to_anchor=(1.05, -0.60),
    ncol=2)

plt.tight_layout()
plt.show()
```



➡️ Use Groupby,pivot tables and correlation analysis

✅ GroupBy

Average loan amount and default rate per loan purpose

In [ ]:
```
#groupby() is used to split data into groups based on categories, and then apply functions (like mean, median, sum, count).
```

In [65]:
```
#summarizing average loan amount and default rate per loan purpose.
#For each Loan Purpose , what is the average LoanAmount for Default=0 vs Default=1 borrowers.
# Group by Loan Purpose and Default
grouped = df.groupby(["LoanPurpose", "Default"])["LoanAmount"].mean().reset_index()
#df.groupby(["LoanPurpose", "Default"])->This groups the dataset  by two categorical columns:LoanPurpose and Default (0 = No D
#["LoanAmount"]->After grouping, needed to perform calculations only on LoanAmount.
#.mean()->For each group (LoanPurpose + Default), calculate the average LoanAmount.
print(grouped.head(10))  # show first 10 rows
```

```
  LoanPurpose  Default     LoanAmount
0        auto        0  125500.978730
1        auto        1  145249.037425
2    business        0  124899.122100
3    business        1  143216.067978
4   education        0  125559.976358
5   education        1  143190.173582
6        home        0  125672.805375
7        home        1  144663.505655
8       other        0  125392.176971
9       other        1  144318.236535
```

Median Income by Education & Default

In [66]:
```
#checks if more educated borrowers have high income,and fewer defaults.
grouped_income = df.groupby(["Education", "Default"])["Income"].median().reset_index() # median provides more realistic centra
print(grouped_income.head(10))
```

```
     Education  Default   Income
0   Bachelor'S        0  82484.0
1   Bachelor'S        1  68664.0
2  High School        0  82583.0
3  High School        1  68308.5
4     Master'S        0  83230.5
5     Master'S        1  69044.0
6          Phd        0  82484.0
7          Phd        1  68438.0
```

Average Debt-to-Income ratio(DTI) by MaritalStatus

In [67]:
```
#Shows whether single/married/divorced borrowers have higher debt burdens.
grouped_dti = df.groupby("MaritalStatus")["DTIRatio"].mean().sort_values(ascending=False)
print(grouped_dti)
```

```
MaritalStatus
Single      0.501448
Married     0.499850
Divorced    0.498966
Name: DTIRatio, dtype: float64
```

✅ Pivot Tables

In [ ]:
```
#A pivot table is like a more flexible version of groupby().It summarizes data in a matrix format-> rows × columns
```

LoanAmount by Education & Marital Status

In [68]:
```
#This pivot table reveals how education & marital status influence loan amounts.
pivot3 = pd.pivot_table(
    df,
    values="LoanAmount",
    index="Education",  #rows
    columns="MaritalStatus",
    aggfunc="mean" # shows average loan amount
)
print(pivot3)
```

```
MaritalStatus       Divorced        Married         Single
Education
Bachelor'S     128347.101148  126582.768679  127280.502347
High School    126807.189658  127008.367468  128377.320639
Master'S       127625.472467  127635.264467  127858.311714
Phd            128443.825416  127822.075225  127177.052288
```

Default Rate by Education & LoanPurpose

In [69]:
```
#checks which education groups default more often for each loan purpose.
pivot4 = pd.pivot_table(
    df,
```

```
    values="Default",
    index="Education",
    columns="LoanPurpose",
    aggfunc="mean"
)
print(pivot4)
```

```
LoanPurpose      auto  business  education      home     other
Education
Bachelor'S   0.120239  0.129342   0.124036  0.106007  0.124595
High School  0.131442  0.137676   0.128670  0.119591  0.128583
Master'S     0.111120  0.114835   0.113083  0.093888  0.111034
Phd          0.112872  0.109529   0.106504  0.090361  0.107741
```

Income vs LoanPurpose

In [70]:
```
#for each loan purpose, what will be typical median borrower income.
pivot5 = pd.pivot_table(
    df,
    values="Income",
    index="LoanPurpose",
    aggfunc="median"
)
print(pivot5)
```

```
              Income
LoanPurpose
auto         82484.0
business     82484.0
education    82484.0
home         82484.0
other        82484.0
```

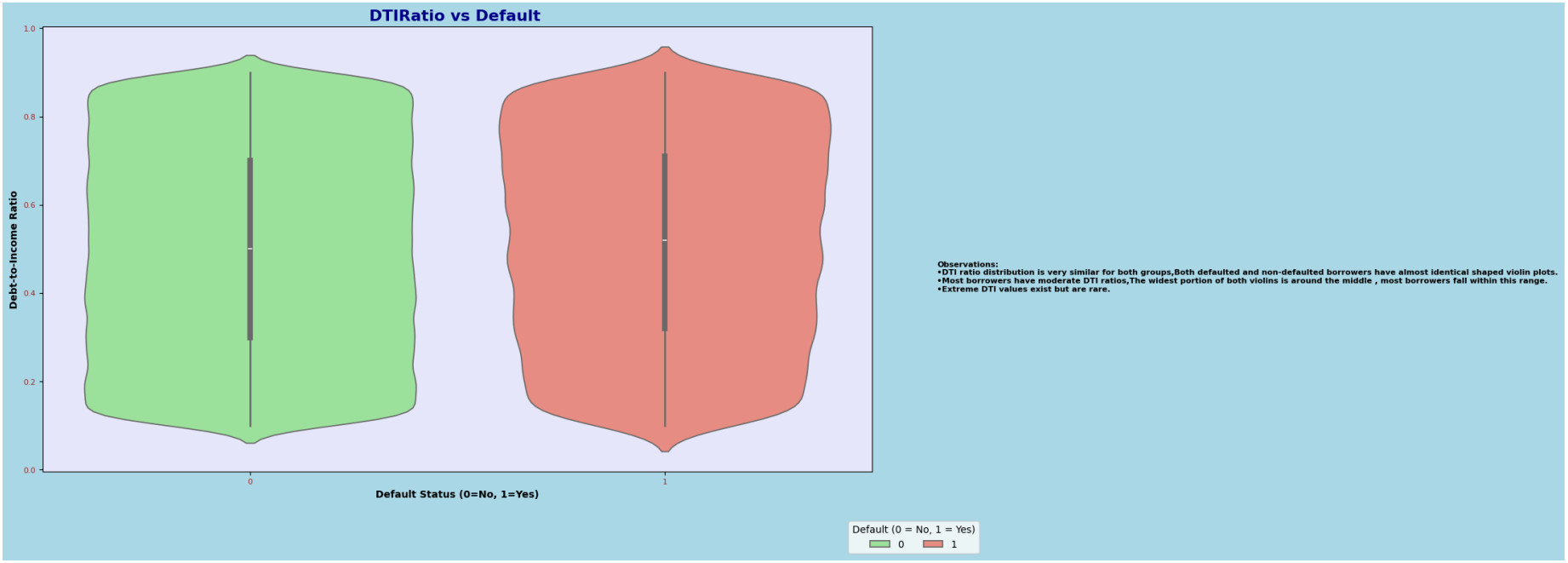
✅ Correlation Analysis

In [ ]:
```
# Correlation analysis measures the strength and direction of the linear relationship between two numerical variables.
```

DTI Ratio vs Default

In [89]:
```
#This analysis is done to check if DTI ratio is an important risk indicator for loan default.
dti_default_corr = df["DTIRatio"].corr(df["Default"])
```

```
plt.figure(figsize=(14,8), facecolor="lightblue")

sns.violinplot(x="Default", y="DTIRatio", data=df,hue="Default", palette={0:"lightgreen",1:"salmon"})

plt.title("DTIRatio vs Default ", #dti_default_corr:2f->variable showing the correlation value computed to 2 decimal values.
          fontsize=16, weight="bold", color="darkblue")

plt.xlabel("Default Status (0=No, 1=Yes)", fontsize=10, weight="bold")
plt.ylabel("Debt-to-Income Ratio", fontsize=10, weight="bold")

plt.xticks(color="brown", fontsize=8)
plt.yticks(color="brown", fontsize=8)

plt.gca().set_facecolor("lavender")

plt.figtext(0.95,0.5,
            "Observations:\n"
            "•DTI ratio distribution is very similar for both groups,Both defaulted and non-defaulted borrowers have almost id
            "•Most borrowers have moderate DTI ratios,The widest portion of both violins is around the middle , most borrowers
            "•Extreme DTI values exist but are rare.\n",
            ha="left", va="center", fontsize=8, weight="bold", color="black")

plt.legend(title="Default (0 = No, 1 = Yes)",
           loc="upper center", bbox_to_anchor=(1.05, -0.10), ncol=2)

plt.tight_layout()
plt.show()
```



Heatmap with colour intensity

In [90]:
```
#This analysis is used to understand what increases or decreases the chance of loan default and to decide which features are u

#A heatmap is a graphical way of showing data values where numbers are represented by colors.
#->Darker / brighter colors = stronger values
#->Lighter / pale colors = weaker values
# Keep only numeric columns

numeric_df = df.select_dtypes(include=["int64","float64"]) # correlation calculates only numbers,so numerical columns with int

# Create heatmap
plt.figure(figsize=(10,7), facecolor="lightblue")
sns.heatmap(numeric_df.corr(), cmap="YlGnBu", cbar=True, annot=False, linewidths=0.5, linecolor="white")

#cmap="YlGnBu" → color scheme (Yellow = low correlation, Blue = high correlation).
#cbar=True → adds a color scale bar on the side.
#annot=False → don't display the actual correlation numbers inside cells (only color intensity).
#linewidths=0.5, linecolor="white" → adds white grid lines to separate cells.
```

```
plt.title("Correlation Heatmap (Color Intensity Only)", fontsize=16, weight="bold", color="darkblue")

plt.gca().set_facecolor("lavender")


plt.figtext(1.02, 0.5,
            "Observations:\n"
            "•Strong positive correlations-LoanAmount and Income show a clear positive correlation → higher income borrowers t
            "•Moderate correlations-Age and Years_Employed have a moderate positive relationship (older borrowers tend to have
            "•Weak or no correlation-Default does not show any strong correlation with a single variable → meaning loan defaul
            "•Negative correlations-A few light-yellow areas indicate negative correlations, but they appear weak , not very s
            ha="left", va="center", fontsize=9, weight="bold", color="black")

plt.tight_layout()
plt.show()
```

➡️ Include Statistical summaries to support findings

In [91]:
```
df.describe()
```

Out[91]:

	Age	Income	LoanAmount	CreditScore	MonthsEmployed	NumCreditLines	InterestRate	LoanTerm	DT
count	261731.0	261731.000000	261731.000000	261731.000000	261731.000000	261731.000000	261731.000000	261731.000000	261731.(
mean	43.486152	82498.931296	127577.088217	574.257898	59.547948	2.500835	13.495161	36.034662	0.5
std	14.806512	38484.896175	69971.416221	156.953952	34.640943	1.117029	6.636820	16.971727	0.2
min	18.0	15000.000000	5000.000000	300.000000	0.000000	1.000000	2.000000	12.000000	0.1
25%	31.0	49638.000000	67645.000000	440.000000	30.000000	2.000000	7.770000	24.000000	0.3
50%	43.0	82484.000000	127506.000000	574.000000	60.000000	2.000000	13.460000	36.000000	0.5
75%	56.0	115394.500000	187464.000000	708.000000	90.000000	3.000000	19.250000	48.000000	0.7
max	69.0	149999.000000	249999.000000	849.000000	119.000000	4.000000	25.000000	60.000000	0.9

4. Visualisations

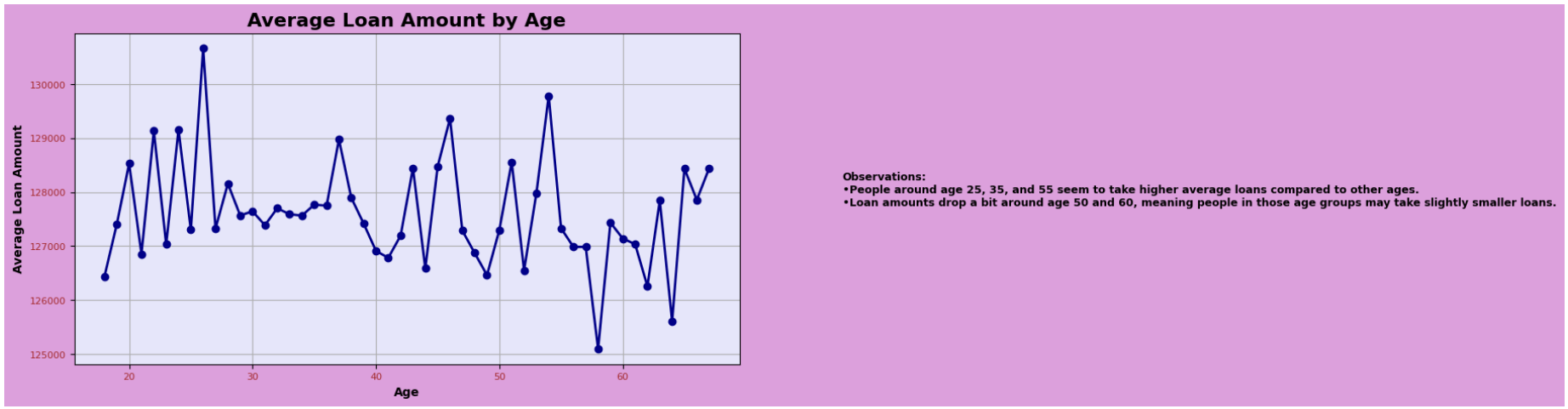
➡️ Use Matplotlib/Seaborn/Plotly to generate meaningful visualisations:

Average LoanAmount by Age

In [16]:

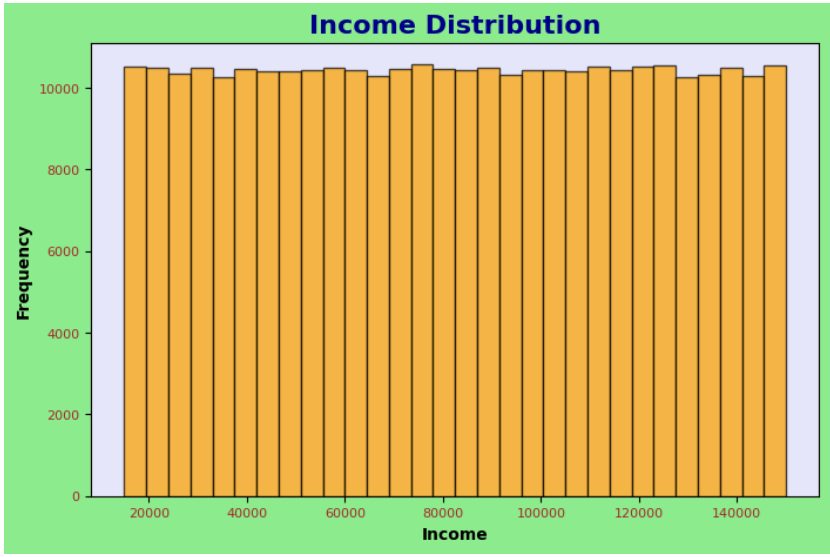
```
#Taking average, how much loan do people of each age take.
# Group data by 'Age' and calculate average loan amount
# This will give  the mean loan amount for each unique age in the dataset.
avg_loan_by_age = df.groupby("Age")["LoanAmount"].mean()


# Sometimes, there are too many unique ages, so we only take the first 50 for neat plotting.
avg_loan_by_age = avg_loan_by_age.head(50)

 #Create the line chart
plt.figure(figsize=(10,5),facecolor="plum")
avg_loan_by_age.plot(                                    # plot a line chart
```

```
    kind="line",                                        # type of chart
    marker="o",                                         # add circular markers on each point
    linewidth=2,                                        # thickness of the line
    color="darkblue"
)


plt.title("Average Loan Amount by Age", fontsize=16, weight="bold")
plt.gca().set_facecolor("lavender")
plt.xlabel("Age",fontsize=10,color="black",weight="bold")
plt.ylabel("Average Loan Amount",fontsize=10,color="black",weight="bold")
plt.xticks(color="brown", fontsize=8)
plt.yticks(color="brown", fontsize=8)

plt.figtext(1.02, 0.5,
            "Observations:\n"
             "•People around age 25, 35, and 55 seem to take higher average loans compared to other ages.\n"
            "•Loan amounts drop a bit around age 50 and 60, meaning people in those age groups may take slightly smaller loans
            ha="left", va="center", fontsize=9, weight="bold", color="black")

plt.grid(True)


plt.show()
```



Income Distribution

In [20]:
```
#how incomes are spread out (who earns less, who earns more, and where most people lie).
# Create histogram of Income column
plt.figure(figsize=(8,5),facecolor="lightgreen")
plt.hist(
    df["Income"],                                          # the column we want to plot
    bins=30,                                               # number of bars (intervals)
    color="orange",
    edgecolor="black",
    alpha=0.7                                              # transparency (0=transparent, 1=solid)
)


plt.title("Income Distribution",color="darkblue", fontsize=16, weight="bold")
plt.gca().set_facecolor("lavender")
plt.xlabel("Income",color="black",weight="bold",fontsize="10")
plt.ylabel("Frequency",color="black",weight="bold",fontsize="10")
plt.xticks(color="brown",fontsize=8)
plt.yticks(color="brown",fontsize=8)
plt.figtext(1.02, 0.5,
            "Observations:\n"
             "•Most people have similar income levels – the bars are almost the same height across all income ranges.\n"
            "•Income ranges from around ₹20,000 to ₹140,000, and in each range, the number of people is nearly equal.\n"
            "•This means that income is evenly distributed in the dataset.\n",
            ha="left", va="center", fontsize=9, weight="bold", color="black")



plt.show()
```

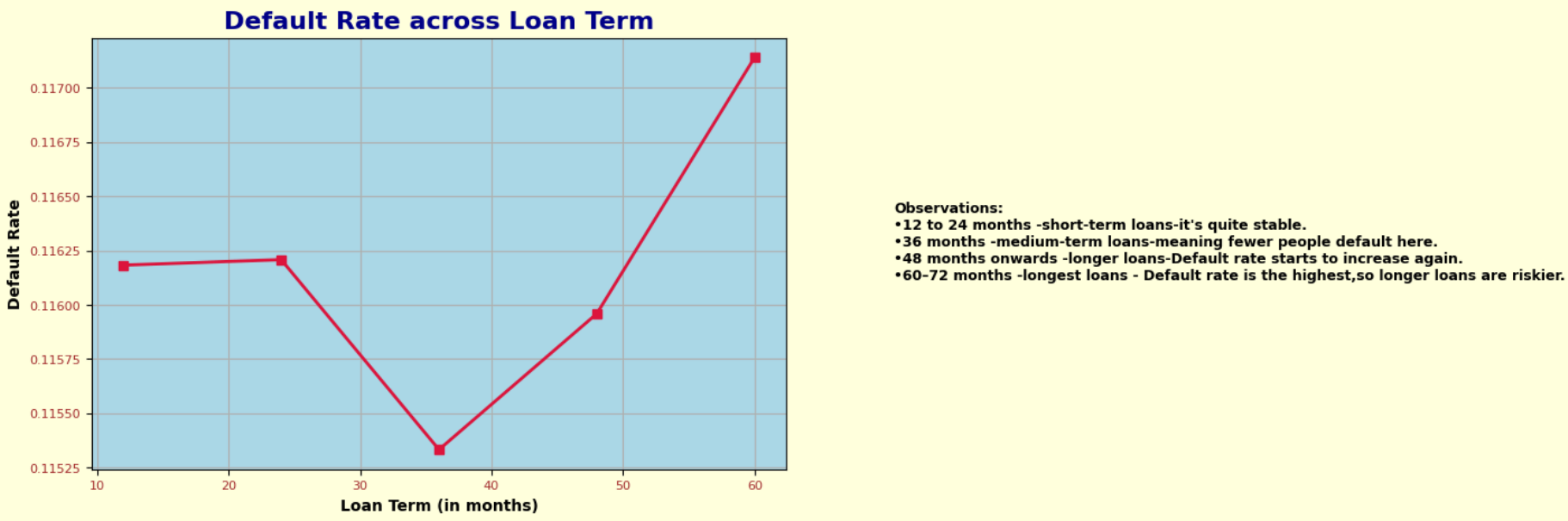


Observations:
- **Most people have similar income levels - the bars are almost the same height across all income ranges.**
- **Income ranges from around ₹20,000 to ₹140,000, and in each range, the number of people is nearly equal.**
- **This means that income is evenly distributed in the dataset.**

Default rate by Loan term

In [26]:
```
#checks if  the loanterm is longer or shorter, does people default more or less.

# Group data by LoanTerm and calculate default rate
# Default is 0 or 1 , so the mean gives us the % of defaults.
default_rate_term = df.groupby("LoanTerm")["Default"].mean()

#  Create the line chart
plt.figure(figsize=(8,5),facecolor="lightyellow")
default_rate_term.plot(
    kind="line",
    marker="s",
    linewidth=2,
    color="crimson"
)

plt.title("Default Rate across Loan Term", fontsize=16,color="darkblue", weight="bold")
plt.gca().set_facecolor("lightblue")
plt.xlabel("Loan Term (in months)",fontsize=10,weight="bold")
plt.ylabel("Default Rate",fontsize=10,weight="bold")
plt.xticks(color="brown",fontsize=8)
plt.yticks(color="brown",fontsize=8)
```

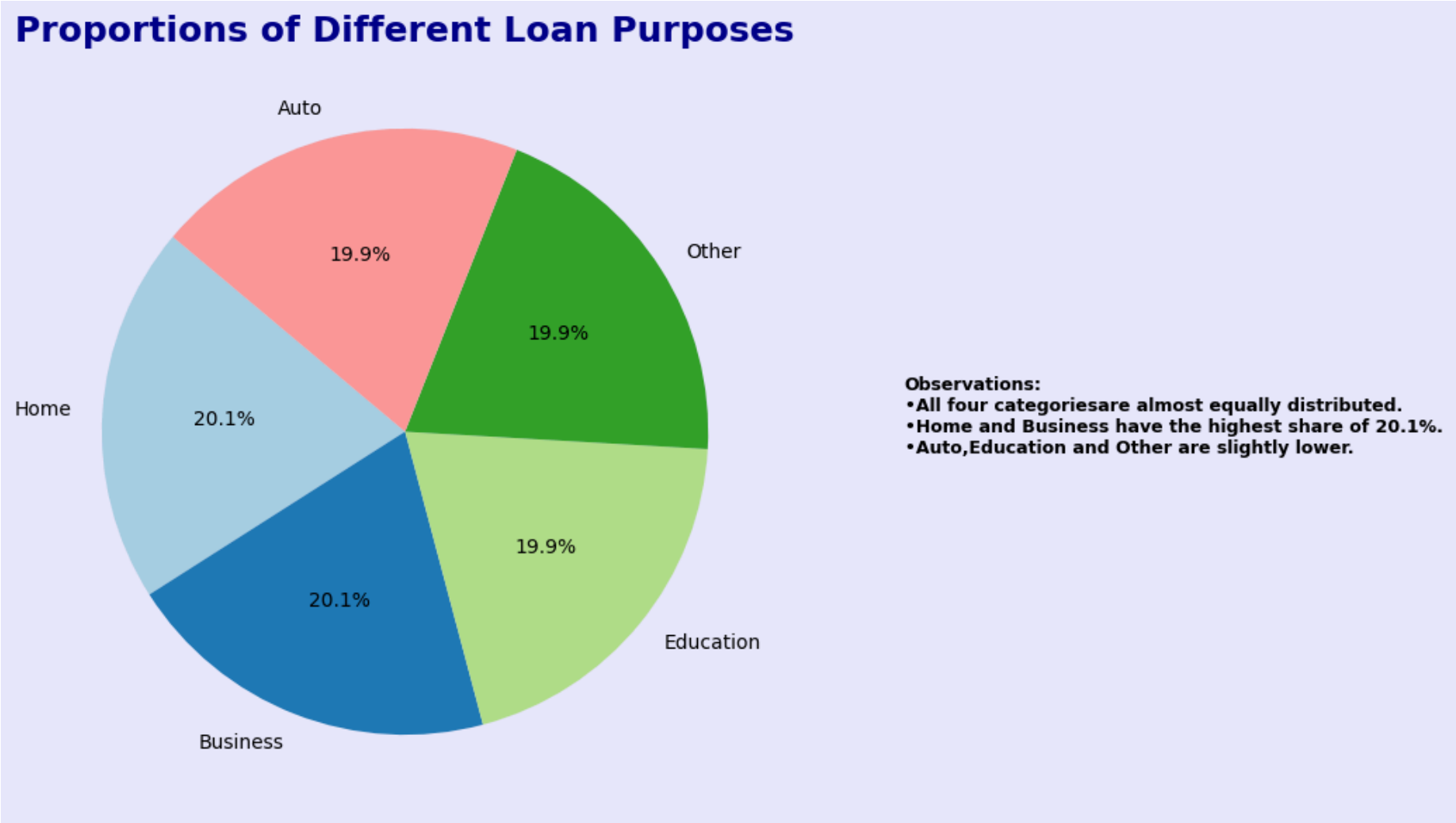
```
plt.figtext(1.02, 0.5,
            "Observations:\n"
            "•12 to 24 months -short-term loans-it's quite stable.\n"
            "•36 months -medium-term loans-meaning fewer people default here.\n"
            "•48 months onwards -longer loans-Default rate starts to increase again.\n"
            "•60–72 months -longest loans - Default rate is the highest,so longer loans are riskier.\n",
            ha="left", va="center", fontsize=9, weight="bold", color="black")



plt.grid(True)


plt.show()
```



Loan Purpose

In [10]:
```
# Count the frequency of each Loan Purpose
loanpurpose_counts = df["LoanPurpose"].value_counts()

# Create the pie chart
```

```
plt.figure(figsize=(7,7),facecolor="lavender")
plt.pie(loanpurpose_counts,
        labels=loanpurpose_counts.index,                      # categories (loan purposes)
        autopct="%1.1f%%",                                   # show percentages
        startangle=140,
        colors=plt.cm.Paired.colors)

plt.title("Proportions of Different Loan Purposes",color="darkblue", fontsize=18, weight="bold")
plt.figtext(1.02, 0.5,
            "Observations:\n"
            "•All four categoriesare almost equally distributed.\n"
            "•Home and Business have the highest share of 20.1%.\n"
            "•Auto,Education and Other are slightly lower.\n",
            ha="left", va="center", fontsize=9, weight="bold", color="black")



plt.show()
```



🔹 Conclusions

- People with low credit scores are more likely to default.

- Borrowers who take a big loan compared to their income default more often.
- Borrowers with stable jobs (longer employment) are less likely to default.
- Having too many credit lines increases the chance of default.
- Younger borrowers (20–30 age group) tend to default more often than older borrowers.
- Defaults cluster in lower income groups → borrowers with small salaries but high loan requests struggle to repay.
- High risk borrowers → young, new to jobs, low income, low credit score, many loans, risky loan purpose
- Low risk borrowers → older, stable jobs, good income, high credit score, fewer loans, productive loan purpose.

◆ Recommendations

- Provide loans mainly to people with good credit scores (above 650).
- Avoid approving loans that are too large compared to income.
- Mostly Prefer borrowers with stable jobs or ask new employees to get a co-signer.
- Make sure to check how many loans/credit lines a borrower already has before giving a new one.
- Apply stricter rules for younger borrowers like smaller loans, co-signer.
- Encourage borrowers with short employment history to build stability before taking big loans.
- Create income-based loan limit,by not allowing to take loans more than a certain percentage of income.
- Use Credit Score & Loan-to-Income ratio as the main approval criteria.
- Build a risk score system by combining credit score, income, loan purpose, employment length and then approve loans only if the risk score is acceptable.

The End

In [ ]:

```

```